



Triebel–Lizorkin spaces in Dunkl setting

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Abstract

We establish Triebel–Lizorkin spaces in the Dunkl setting which are associated with finite reflection groups on the Euclidean space. The group structures induce two non-equivalent metrics: the Euclidean metric and the Dunkl metric. In this paper, the L^2 space and the Dunkl–Calderón–Zygmund singular integral operator in the Dunkl setting play a fundamental role. The main tools used in this paper are as follows: (i) the Dunkl–Calderón–Zygmund singular integral operator and a new Calderón reproducing formula in L^2 with the Triebel–Lizorkin space norms; (ii) new test functions in terms of the L^2 functions and distributions; (iii) the Triebel–Lizorkin spaces in the Dunkl setting which are defined by the wavelet-type decomposition and the analogous atomic decomposition of the Hardy spaces.

Keywords Triebel–Lizorkin spaces · Dunkl setting · Calderón reproducing formula · Test functions · Distributions

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1 Introduction

It is well-known that the Triebel–Lizorkin spaces play an important role in modern harmonic analysis. In the 1970s, the Triebel–Lizorkin space $\dot{F}_p^{\alpha,q}$ on $\mathbb{R}^n$ was introduced by several mathematicians. Lizorkin [13, 14] and Triebel [20] independently investigated the Triebel–Lizorkin space $\dot{F}_p^{\alpha,q}(\mathbb{R}^n)$, $\alpha \in \mathbb{R}$, $1 < p < \infty$, $1 < q < \infty$. Peetre [15–17] extended the range of the admissible parameters p

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and q to values less than one. In [7], applying the basic representation formula of the form $f = \sum_Q \langle f, \varphi_Q \rangle \psi_Q$ for a distribution f on $\mathbb{R}^n$, the Triebel–Lizorkin space was defined through the Littlewood–Paley theory. More precisely, let φ, ψ be functions on $\mathbb{R}^n$ satisfying:

$$\begin{aligned} \varphi, \psi &\in \mathcal{S}(\mathbb{R}^n), \\ \text{supp } \hat{\varphi}, \text{supp } \hat{\psi} &\subseteq \left\{ \xi \in \mathbb{R}^n : \frac{1}{2} \leq |\xi| \leq 2 \right\}, \\ |\hat{\varphi}(\xi)|, |\hat{\psi}(\xi)| &\geq c > 0 \quad \text{if } \frac{3}{5} \leq |\xi| \leq \frac{5}{3}, \end{aligned}$$

and

$$\sum_{k \in \mathbb{Z}} \overline{\hat{\varphi}(2^k \xi)} \hat{\psi}(2^k \xi) = 1 \quad \text{if } \xi \neq 0.$$

Set $\varphi_k(x) = 2^{kn} \varphi(2^k x)$ and $\psi_k(x) = 2^{kn} \psi(2^k x)$, $k \in \mathbb{Z}$. The Triebel–Lizorkin space $\dot{F}_p^{\alpha, q}(\mathbb{R}^n)$, $\alpha \in \mathbb{R}$, $0 < p < \infty$, $0 < q \leq \infty$, was defined by the collection of all $f \in \mathcal{S}' / \mathcal{P}(\mathbb{R}^n)$ (tempered distributions modulo polynomials) such that

$$\|f\|_{\dot{F}_p^{\alpha, q}} = \left\| \left(\sum_{k \in \mathbb{Z}} (2^{k\alpha} |\varphi_k * f|)^q \right)^{\frac{1}{q}} \right\|_{L^p(\mathbb{R}^n)} < \infty,$$

$$\text{and when } q = \infty, \|f\|_{\dot{F}_p^{\alpha, \infty}} = \left\| \sup_{k \in \mathbb{Z}} 2^{k\alpha} |\varphi_k * f(x)| \right\|_{L^p(\mathbb{R}^n)}.$$

As a generalization of $\mathbb{R}^n$, the space of homogeneous type was introduced by Coifman and Weiss in [3], which provides a natural setting for studying function spaces. The homogeneous Triebel–Lizorkin spaces on spaces of homogeneous type were studied in [9–12]. More precisely, the Triebel–Lizorkin spaces on homogeneous type were introduced by using the family of operators $\{\mathbf{D}_k\}_{k \in \mathbb{Z}}$ where $\mathbf{D}_k = \mathbf{S}_k - \mathbf{S}_{k-1}$ and $\{\mathbf{S}_k\}_{k \in \mathbb{Z}}$ is an approximation to the identity. For $|\alpha| < \varepsilon$, $1 < p, q < \infty$, the Triebel–Lizorkin space $\dot{F}_p^{\alpha, q}$ is the collection of all $f \in (\mathcal{M}^{(\beta, \gamma)})'$ with $0 < \beta, \gamma < \varepsilon$ such that

$$\|f\|_{\dot{F}_p^{\alpha, q}} = \left\| \left\{ \sum_{k \in \mathbb{Z}} (2^{k\alpha} |\mathbf{D}_k(f)|)^q \right\}^{\frac{1}{q}} \right\|_{L^p(\mathbb{R}^n)} < \infty,$$

where $\mathcal{M}^{(\beta, \gamma)}$ is the space of test functions. See [9–12] for more details.

In recent years, the Dunkl setting is more and more important. In [10], the authors introduced the Triebel–Lizorkin spaces $\dot{F}_p^{\alpha, q}$ in the Dunkl setting for $1 < p, q < \infty$. In this paper, we extend the range of p, q to all $0 < p, q < \infty$. The key tool is the discrete Calderón reproducing formula derived from the Dunkl–Poisson kernel (see [18]). We mention that the Dunkl–Poisson kernel involves two nonequivalent metrics: the Euclidean metric and the Dunkl metric.

Now we recall the framework of the Dunkl setting, see [5, 6, 19]. In $\mathbb{R}^n$, the reflection σ_α with respect to the hyperplane $\alpha_\perp$ orthogonal to a nonzero vector α is given by

$$\sigma_\alpha(x) = x - 2 \frac{\langle x, \alpha \rangle}{\|\alpha\|^2} \alpha.$$

A finite set $R \subset \mathbb{R}^n \setminus \{0\}$ is called a root system if $\sigma_\alpha(R) = R$ for every $\alpha \in R$. Let R be a root system in $\mathbb{R}^n$ normalized so that $\langle \alpha, \alpha \rangle = 2$ for $\alpha \in R$ and G the finite reflection group generated by the reflections σ_α ($\alpha \in R$), where $\sigma_\alpha(x) = x - \langle \alpha, x \rangle \alpha$ for $x \in \mathbb{R}^n$. Corresponding to this reflection group, we denote by $\mathcal{O}(x)$ the G -orbit of a point $x \in \mathbb{R}^n$. There is a natural metric between two G -orbits $\mathcal{O}(x)$ and $\mathcal{O}(y)$, given by

$$d(x, y) := \min_{\sigma \in G} \|x - \sigma(y)\|. \quad (1.1)$$

Obviously, $d(x, y) \leq \|x - y\|$, $d(x, y) = d(y, x)$ and $d(x, y) \leq d(x, z) + d(z, y)$ for all $x, y, z \in \mathbb{R}^n$.

A multiplicity function κ defined on R (invariant under G) is fixed throughout this paper. Let

$$d\omega(x) = \prod_{\alpha \in R} |\langle \alpha, x \rangle|^{\kappa(\alpha)} dx$$

be the associated measure in $\mathbb{R}^n$, see [1], where, here and subsequently, dx stands for the Lebesgue measure in $\mathbb{R}^n$. We denote by $N = n + \sum_{\alpha \in R} \kappa(\alpha)$ the homogeneous dimension of the system, here N is also called the upper dimension. Observe that for $x \in \mathbb{R}^n$ and $r > 0$,

$$\omega(B(x, r)) \sim r^N \prod_{\alpha \in R} (|\langle \alpha, x \rangle| + r)^{\kappa(\alpha)}, \quad (1.2)$$

hence $\inf_{x \in \mathbb{R}^n} \omega(B(x, 1)) \geq C > 0$, and $\omega(B(x, r)) \geq Cr^N$. According to (1.1), we also have $\omega(B(x, r)) \sim \omega(B(y, r))$ when $d(x, y) \sim r$, and $\omega(B(x, r)) \leq \omega(B_d(x, r)) \leq |G|\omega(B(x, r))$, where $B_d(x, r) := \{y \in \mathbb{R}^n : d(x, y) < r\}$.

Moreover, the measure $d\omega(x)$ satisfies

$$C^{-1} \left(\frac{r_2}{r_1} \right)^N \leq \frac{\omega(B(x, r_2))}{\omega(B(x, r_1))} \leq C \left(\frac{r_2}{r_1} \right)^N \quad \text{for } 0 < r_1 < r_2.$$

This implies that $d\omega(x)$ satisfies the doubling and reverse doubling properties, that is, there exists a constant $C > 0$ such that for all $x \in \mathbb{R}^n$, $r > 0$ and $\lambda \geq 1$,

$$C^{-1} \lambda^N \omega(B(x, r)) \leq \omega(B(x, \lambda r)) \leq C \lambda^N \omega(B(x, r)).$$

Next we consider the Dunkl (differential) operators $\mathbf{T}_j$ defined by

$$\mathbf{T}_j f(x) = \partial_j f(x) + \sum_{\alpha \in R} \frac{\kappa(\alpha)}{2} \langle \alpha, e_j \rangle \frac{f(x) - f(\sigma_\alpha(x))}{\langle \alpha, x \rangle},$$

where $e_1, e_2, \dots, e_n$ are standard unit vectors of $\mathbb{R}^n$. The Dunkl Laplacian is then defined as $\Delta_D = \sum_{j=1}^n \mathbf{T}_j^2$, which is equivalent to

$$\Delta_D f(x) = \Delta_{\mathbb{R}^n} f(x) + \sum_{\alpha \in R} \kappa(\alpha) \left(\frac{\partial_\alpha f(x)}{\langle \alpha, x \rangle} - \frac{f(x) - f(\sigma_\alpha(x))}{\langle \alpha, x \rangle^2} \right).$$

Here $\Delta_{\mathbb{R}^n}$ is the standard Euclidean Laplacian.

The operator Δ_D is self-adjoint on $L^2(\mathbb{R}^n, \omega)$, see [1], and generates the heat semigroup

$$\mathbf{H}_t f(x) = e^{t\Delta_D} f(x) = \int_{\mathbb{R}^n} H_t(x, y) f(y) d\omega(y),$$

where the heat kernel $H_t(x, y)$ is a C^∞ function for all $t > 0$, $x, y \in \mathbb{R}^n$ and satisfies $H_t(x, y) = H_t(y, x) > 0$ and $\int_{\mathbb{R}^n} H_t(x, y) d\omega(y) = 1$.

The Poisson semigroup $\mathbf{P}_t = e^{-t\sqrt{-\Delta_D}}$ is subordinated to the heat semigroup $\mathbf{H}_t = e^{t\Delta_D}$ by

$$\mathbf{P}_t f(x) = \pi^{-1/2} \int_0^\infty e^{-u} e^{\frac{t^2}{4u} \Delta_D} f(x) \frac{du}{\sqrt{u}},$$

and correspondingly for their integral kernels

$$P_t(x, y) = \pi^{-1/2} \int_0^{2\pi} e^{-u} H_{\frac{t^2}{4u}}(x, y) \frac{du}{u}.$$

Moreover, $u(x, t) = \mathbf{P}_t f(x)$, so-called the Dunkl–Poisson integral, solves the boundary value problem

$$\begin{cases} (\partial_t^2 + \Delta_D)u(x, t) = 0, \\ u(x, 0) = f(x), \end{cases}$$

in the half-space $\mathbb{R}_+^{n+1}$, see [1].

Observe that $\{\mathbf{P}_{2^{-k}}\}_{k \in \mathbb{Z}}$ is an approximation to the identity on $L^2(\mathbb{R}^n, \omega)$, that is, for $f \in L^2(\mathbb{R}^n, \omega)$

$$\lim_{k \rightarrow +\infty} \mathbf{P}_{2^{-k}}(f)(x) = f(x), \quad \lim_{k \rightarrow -\infty} \mathbf{P}_{2^{-k}}(f)(x) = 0.$$

◊ Throughout, unless otherwise specified, boldface letters denote operators, while the corresponding plain letters denote their kernels. For example,

$$\mathbf{D}_k(f)(x) = \int_{\mathbb{R}^n} D_k(x, y) f(y) d\omega(y).$$

Now we consider the Triebel–Lizorkin spaces in the Dunkl setting. In [2], by using the Dunkl Laplacian Δ_D , the author proved that the Triebel–Lizorkin spaces associated with the Dunkl–Laplacian are identical to the Triebel–Lizorkin spaces defined in the space of homogeneous type $(\mathbb{R}^n, \|\cdot\|, \omega)$.

In this paper, we establish the Triebel–Lizorkin spaces in the Dunkl setting in a different way. The critical difference is we use the Calderón reproducing formula derived from two different kernels. Our new method is using a new Calderón reproducing formula in L^2 (see [18]) with the Triebel–Lizorkin space norms derived from the Dunkl–Poisson kernel. Set

$$\mathbf{D}_k = \mathbf{P}_{2^{-k}} - \mathbf{P}_{2^{-k-1}}$$

with the kernel $D_k(x, y) = P_{2^{-k}}(x, y) - P_{2^{-k-1}}(x, y)$. Then applying the Coifman’s decomposition (see [4]) of the identity on $L^2(\mathbb{R}^n, \omega)$, we have

$$\mathbf{I} = \left(\sum_{\ell \in \mathbb{Z}} \mathbf{D}_\ell \right) \left(\sum_{k \in \mathbb{Z}} \mathbf{D}_k \right) = \sum_{|k-\ell| \leq M} \mathbf{D}_\ell \mathbf{D}_k + \sum_{|k-\ell| > M} \mathbf{D}_\ell \mathbf{D}_k = \sum_{k \in \mathbb{Z}} \mathbf{D}_k^M \mathbf{D}_k + \mathbf{R}_1$$

where M is a fixed constant, and

$$\mathbf{D}_k^M = \sum_{\ell: |k-\ell| \leq M} \mathbf{D}_\ell, \mathbf{R}_1 = \sum_{|k-\ell| > M} \mathbf{D}_\ell \mathbf{D}_k.$$

Therefore, for $f \in L^2(\mathbb{R}^n, \omega)$ we have

$$f(x) = \sum_{k \in \mathbb{Z}} \mathbf{D}_k^M \mathbf{D}_k(f)(x) + \mathbf{R}_1(f)(x) = \mathbf{T}_M(f)(x) + \mathbf{R}_1(f)(x) + \mathbf{R}_2(f)(x),$$

where

$$\begin{aligned} \mathbf{T}_M(f)(x) &= \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(f)(x_Q), \\ \mathbf{R}_2(f)(x) &= \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(\int_Q D_k^M(x, y) \mathbf{D}_k(f)(y) d\omega(y) - \omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(f)(x_Q) \right), \end{aligned}$$

where Q_d^k is the collection of all dyadic cubes Q with the side length 2^{-M-k} , M is some fixed large integer, and x_Q is any fixed point in the cube Q .

By showing the operator $\mathbf{T}_M$ is invertible, the authors in [18] proved the Calderón reproducing formula on L^2 as following

Theorem 1.1 [18] *If $f \in L^2(\mathbb{R}^n, \omega)$, then there exists a function $h \in L^2(\mathbb{R}^n, \omega)$, such that $\|f\|_2 \sim \|h\|_2$ and*

$$f(x) = \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(h)(x_Q)$$

where Q are dyadic cubes in $\mathbb{R}^n$, Q_d^k is the collection of Q with the side length $\ell(Q) = 2^{-k-M}$, and x_Q are any fixed point in Q .

Based on the above theorem, we introduce the following

Definition 1.2 Suppose that $f \in L^2(\mathbb{R}^n, \omega)$, $\alpha \in \mathbb{R}$, $0 < p < \infty$, $0 < q < \infty$, the Littlewood–Paley q -function $\mathbf{S}_q^\alpha(f)$ is defined by

$$\mathbf{S}_q^\alpha(f)(x) := \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{k\alpha} |\mathbf{D}_k(f)(x_Q)| \right)^q \chi_Q(x) \right\}^{\frac{1}{q}},$$

where $\chi_Q(x)$ is the characteristic function of the cube Q .

To establish the Triebel–Lizorkin space in the Dunkl setting, we utilize the new Calderón reproducing formula in L^2 with the Triebel–Lizorkin space norms derived from the Dunkl–Poisson kernel (see [1]). Next, we introduce the new test functions in terms of the L^2 functions and distributions by the duality estimates, which is crucial for developing the Dunkl–Triebel–Lizorkin spaces. Then we consider the Dunkl–Triebel–Lizorkin spaces as the collection of some distributions on the test function, and define the spaces by the wavelet-type decomposition and the analogous atomic decomposition of the Hardy spaces. Finally, we prove the Dunkl–Triebel–Lizorkin spaces defined this way are complete.

2 Formulation on the main results

The discrete Littlewood–Paley q -function in Definition 1.2 leads to introduce the Dunkl–Triebel–Lizorkin space norm for $f \in L^2(\mathbb{R}^n, \omega)$ as follows:

Definition 2.1 For $f \in L^2(\mathbb{R}^n, \omega)$, the Dunkl–Triebel–Lizorkin space norm of f is defined as

$$\|f\|_{\dot{F}_p^{\alpha,q}} := \|\mathbf{S}_q^\alpha(f)\|_p$$

for $\alpha \in \mathbb{R}$, $0 < p < \infty$ and $0 < q < \infty$.

Here and subsequently, for $0 < p < \infty$, we denote the norm of $f \in L^p(\mathbb{R}^n, \omega)$ by

$$\|f\|_p := \|f\|_{L^p(\mathbb{R}^n, \omega)} = \left\{ \int_{\mathbb{R}^n} |f(x)|^p d\omega(x) \right\}^{\frac{1}{p}} < \infty.$$

Our first main result, the discrete Calderón reproducing formula for $f \in L^2(\mathbb{R}^n, \omega)$ with respect to the Dunkl–Triebel–Lizorkin space norm, is given by the following

Theorem 2.2 If $f \in L^2(\mathbb{R}^n, \omega)$ with $\|f\|_{\dot{F}_p^{\alpha,q}} < \infty$, for $|\alpha| < 1$, $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p < \infty$, $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q < \infty$, where N is the upper dimension, then there exists a function $h \in L^2(\mathbb{R}^n, \omega)$, such that $\|f\|_2 \sim \|h\|_2$, $\|f\|_{\dot{F}_p^{\alpha,q}} \sim \|h\|_{\dot{F}_p^{\alpha,q}}$ and

$$f(x) = \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(h)(x_Q),$$

where Q_d^k is the collection of all dyadic cubes Q with the side length 2^{-M-k} , M is some fixed large integer, and x_Q is any fixed point in the cube Q . The series converges in $L^2(\mathbb{R}^n, \omega)$ norm and the Dunkl–Triebel–Lizorkin space norm.

Applying the above theorem, we provide the duality estimates which will be a key idea for developing the Dunkl–Triebel–Lizorkin space theory. Before we state the estimates, we first define some necessary space norms.

Definition 2.3 Suppose that $|\alpha| < 1$, $0 < p \leq 1$, $1 < q < \infty$ and $f \in L^2(\mathbb{R}^n, \omega)$. The norm of f in the Dunkl–Carleson measure space $CMO_p^{\alpha, q}$ is defined by

$$\|f\|_{CMO_p^{\alpha, q}} = \sup_P \left(\frac{1}{\omega(P)^{\frac{q}{p} - \frac{q}{q'}}} \sum_{Q \in P} \omega(Q) |2^{k\alpha} \mathbf{D}_k^M(f)(x_Q)|^q \right)^{\frac{1}{q}},$$

where P runs over all dyadic cubes.

Definition 2.4 Suppose that $|\alpha| < 1$, $1 \leq p < \infty$ and $f \in L^2(\mathbb{R}^n, \omega)$. The norm of $f \in \dot{F}_p^{\alpha, \infty}(\mathbb{R}^n, \omega)$ is defined by

$$\|f\|_{\dot{F}_p^{\alpha, \infty}} = \left\| \sup_{k \in \mathbb{Z}, Q \in Q_d^k} 2^{k\alpha} |\mathbf{D}_k^M(f)(x_Q)| \chi_Q(x) \right\|_p.$$

Definition 2.5 Suppose that $|\alpha| < 1$ and $f \in L^2(\mathbb{R}^n, \omega)$. The norm of $f \in \dot{F}_\infty^{\alpha, \infty}(\mathbb{R}^n, \omega)$ is defined by

$$\|f\|_{\dot{F}_\infty^{\alpha, \infty}} = \sup_{k \in \mathbb{Z}, Q \in Q_d^k} 2^{k\alpha} |\mathbf{D}_k^M(f)(x_Q)|.$$

Now we return to state the duality estimates

Theorem 2.6 Suppose that $f, g \in L^2(\mathbb{R}^n, \omega)$, p', q' are the conjugates of $1 < p, q < \infty$, respectively.

(A) For $1 < p < \infty, 1 < q < \infty$, there exists a constant C such that

$$|\langle f, g \rangle| \leq C \|f\|_{\dot{F}_p^{\alpha, q}} \|g\|_{\dot{F}_{p'}^{-\alpha, q'}}$$

(B) For $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p \leq 1, 1 < q < \infty$, there exists a constant C such that

$$|\langle f, g \rangle| \leq C \|f\|_{\dot{F}_p^{\alpha, q}} \|g\|_{CMO_p^{-\alpha, q'}}$$

where $\|g\|_{CMO_p^{-\alpha,q'}}$ is defined as in Definition 2.3.

(C) For $1 < p < \infty$, $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q \leq 1$, there exists a constant C such that

$$|\langle f, g \rangle| \leq C \|f\|_{\dot{F}_p^{\alpha,q}} \|g\|_{\dot{F}_{p'}^{-\alpha,\infty}}$$

where $\|g\|_{\dot{F}_{p'}^{-\alpha,\infty}}$ is defined as in Definition 2.4.

(D) For $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p \leq 1$, $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q \leq 1$, there exists a constant C such that

$$|\langle f, g \rangle| \leq C \|f\|_{\dot{F}_p^{\alpha,q}} \|g\|_{\dot{F}_\infty^{-\alpha+N(\frac{1}{p}-1),\infty}}$$

where $\|g\|_{\dot{F}_\infty^{-\alpha+N(\frac{1}{p}-1),\infty}}$ is defined as in Definition 2.5.

The above Theorem 2.6 means that for $1 < p < \infty$, $1 < q < \infty$, each function $f \in L^2(\mathbb{R}^n, \omega)$ with $\|f\|_{\dot{F}_p^{\alpha,q}} < \infty$ can be considered as a continuous linear functional on $L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha,q'}(\mathbb{R}^n, \omega)$, the subspace of $g \in L^2(\mathbb{R}^n, \omega)$ with the norm $\|g\|_{\dot{F}_{p'}^{-\alpha,q'}} < \infty$.

Therefore, one can consider $L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha,q'}(\mathbb{R}^n, \omega)$ as a new test function space and write $f \in \left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{\alpha,q'}(\mathbb{R}^n, \omega) \right)'$, where $\left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{\alpha,q'}(\mathbb{R}^n, \omega) \right)'$ is the distribution space. Other ranges of p, q stated above have the same results. The following result describes an important property for such a distribution f . More precisely, we establish the following discrete Calderón reproducing formula in the distribution sense:

Theorem 2.7 For $|\alpha| < 1$, $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p < \infty$, $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q < \infty$, suppose that $\{f_n\}_{n \in \mathbb{Z}}$ is a Cauchy sequence in $L^2(\mathbb{R}^n, \omega)$ with $\|S_q^\alpha(f_n - f_m)\|_p \rightarrow 0$ as $m, n \rightarrow \infty$. Then there exists a distribution f satisfying

(A) For $1 < p < \infty$, $1 < q < \infty$, f is a distribution in $\left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha,q'}(\mathbb{R}^n, \omega) \right)'$ such that

$$\lim_{n \rightarrow \infty} \|f_n - f\|_{\dot{F}_p^{\alpha,q}} = 0,$$

and for each $g \in L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha,q'}(\mathbb{R}^n, \omega)$,

$$\langle f, g \rangle = \left\langle \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_k^d} \omega(Q) D_k^M(\cdot, x_Q) \mathbf{D}_k(h)(x_Q), g(\cdot) \right\rangle,$$

where $h \in \left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha,q'}(\mathbb{R}^n, \omega) \right)'$ with $\|f\|_{\dot{F}_p^{\alpha,q}} \sim \|h\|_{\dot{F}_p^{\alpha,q}}$;

- (B) For $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p \leq 1$, $1 < q < \infty$, f is a distribution in $\left(L^2(\mathbb{R}^n, \omega) \cap CMO_p^{-\alpha, q'}(\mathbb{R}^n, \omega) \right)'$, such that

$$\lim_{n \rightarrow \infty} \|f_n - f\|_{\dot{F}_p^{\alpha, q}} = 0,$$

and for each $g \in L^2(\mathbb{R}^n, \omega) \cap CMO_p^{-\alpha, q'}(\mathbb{R}^n, \omega)$,

$$\langle f, g \rangle = \left\langle \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(\cdot, x_Q) \mathbf{D}_k(h)(x_Q), g(\cdot) \right\rangle,$$

where $h \in \left(L^2(\mathbb{R}^n, \omega) \cap CMO_p^{-\alpha, q'}(\mathbb{R}^n, \omega) \right)'$ with $\|f\|_{\dot{F}_p^{\alpha, q}} \sim \|h\|_{\dot{F}_p^{\alpha, q}}$;

- (C) For $1 < p < \infty$, $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q \leq 1$, f is a distribution in $\left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha, \infty}(\mathbb{R}^n, \omega) \right)'$ such that

$$\lim_{n \rightarrow \infty} \|f_n - f\|_{\dot{F}_p^{\alpha, q}} = 0,$$

and for each $g \in L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha, \infty}(\mathbb{R}^n, \omega)$,

$$\langle f, g \rangle = \left\langle \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(\cdot, x_Q) \mathbf{D}_k(h)(x_Q), g(\cdot) \right\rangle,$$

where $h \in \left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha, \infty}(\mathbb{R}^n, \omega) \right)'$ with $\|f\|_{\dot{F}_p^{\alpha, q}} \sim \|h\|_{\dot{F}_p^{\alpha, q}}$;

- (D) For $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p \leq 1$, $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q \leq 1$, f is a distribution in $\left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_\infty^{-\alpha + N(\frac{1}{p} - 1), \infty}(\mathbb{R}^n, \omega) \right)'$ such that

$$\lim_{n \rightarrow \infty} \|f_n - f\|_{\dot{F}_p^{\alpha, q}} = 0,$$

and for each $g \in L^2(\mathbb{R}^n, \omega) \cap \dot{F}_\infty^{-\alpha + N(\frac{1}{p} - 1), \infty}(\mathbb{R}^n, \omega)$,

$$\langle f, g \rangle = \left\langle \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(\cdot, x_Q) \mathbf{D}_k(h)(x_Q), g(\cdot) \right\rangle,$$

where $h \in \left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_\infty^{-\alpha + N(\frac{1}{p} - 1), \infty}(\mathbb{R}^n, \omega) \right)'$ with $\|f\|_{\dot{F}_p^{\alpha, q}} \sim \|h\|_{\dot{F}_p^{\alpha, q}}$.

Then we define the Dunkl–Triebel–Lizorkin spaces by the following

Definition 2.8 The Dunkl–Triebel–Lizorkin space $\dot{F}_p^{\alpha, q}(\mathbb{R}^n, \omega)$, $|\alpha| < 1$, $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p < \infty$, $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q < \infty$ is defined as follows:

$$\dot{F}_p^{\alpha,q} = \left\{ f : f(x) = \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) \lambda_Q D_k^M(x, x_Q), \right. \\ \left. \text{with } \left\| \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{k\alpha} |\lambda_Q| \right)^q \chi_Q(x) \right\}^{\frac{1}{q}} \right\|_p < \infty \right\},$$

where the series converges in the following distribution sense:

- (A) for $1 < p < \infty, 1 < q < \infty$, f converges in $\left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha,q'}(\mathbb{R}^n, \omega) \right)'$;
- (B) for $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p \leq 1, 1 < q < \infty$, f converges in $\left(L^2(\mathbb{R}^n, \omega) \cap CMO_p^{-\alpha,q'}(\mathbb{R}^n, \omega) \right)'$;
- (C) for $1 < p < \infty, \max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q \leq 1$, f converges in $\left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha,\infty}(\mathbb{R}^n, \omega) \right)'$;
- (D) for $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p \leq 1, \max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q \leq 1$, f converges in $\left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{\infty}^{-\alpha+N(\frac{1}{p}-1),\infty}(\mathbb{R}^n, \omega) \right)'$.

If $f \in \dot{F}_p^{\alpha,q}(\mathbb{R}^n, \omega)$, the norm of f is defined by

$$\|f\|_{\dot{F}_p^{\alpha,q}} := \inf \left\{ \left\| \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{k\alpha} |\lambda_Q| \right)^q \chi_Q(x) \right\}^{\frac{1}{q}} \right\|_p \right\}$$

where the infimum is taken over all $f(x) = \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) \lambda_Q D_k^M(x, x_Q)$.

To clarify that the Dunkl–Triebel–Lizorkin space in Definition 2.8 is complete, we show the following

Theorem 2.9 For $|\alpha| < 1, \max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p < \infty,$
 $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q < \infty,$

$$\dot{F}_p^{\alpha,q}(\mathbb{R}^n, \omega) = \overline{L^2(\mathbb{R}^n, \omega) \cap \dot{F}_p^{\alpha,q}(\mathbb{R}^n, \omega)}.$$

More precisely,

- (A) for $1 < p < \infty, 1 < q < \infty, \overline{L^2(\mathbb{R}^n, \omega) \cap \dot{F}_p^{\alpha, q}(\mathbb{R}^n, \omega)}$ is the collection of all distributions $f \in \left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha, q'}(\mathbb{R}^n, \omega) \right)'$ such that there exists a sequence $\{f_n\}_{n=1}^\infty$ in $L^2(\mathbb{R}^n, \omega)$ with $\|f_n - f_m\|_{\dot{F}_p^{\alpha, q}} \rightarrow 0$ as $n, m \rightarrow \infty$. Moreover, f_n converges to f in $\left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha, q'}(\mathbb{R}^n, \omega) \right)'$;
- (B) for $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p \leq 1, 1 < q < \infty, \overline{L^2(\mathbb{R}^n, \omega) \cap \dot{F}_p^{\alpha, q}(\mathbb{R}^n, \omega)}$ is the collection of all distributions $f \in \left(L^2(\mathbb{R}^n, \omega) \cap CMO_p^{-\alpha, q'}(\mathbb{R}^n, \omega) \right)'$ such that there exists a sequence $\{f_n\}_{n=1}^\infty$ in $L^2(\mathbb{R}^n, \omega)$ with $\|f_n - f_m\|_{\dot{F}_p^{\alpha, q}} \rightarrow 0$ as $n, m \rightarrow \infty$. Moreover, f_n converges to f in $\left(L^2(\mathbb{R}^n, \omega) \cap CMO_p^{-\alpha, q'}(\mathbb{R}^n, \omega) \right)'$;
- (C) for $1 < p < \infty, \max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q \leq 1, \overline{L^2(\mathbb{R}^n, \omega) \cap \dot{F}_p^{\alpha, q}(\mathbb{R}^n, \omega)}$ is the collection of all distributions $f \in \left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha, \infty}(\mathbb{R}^n, \omega) \right)'$ such that there exists a sequence $\{f_n\}_{n=1}^\infty$ in $L^2(\mathbb{R}^n, \omega)$ with $\|f_n - f_m\|_{\dot{F}_p^{\alpha, q}} \rightarrow 0$ as $n, m \rightarrow \infty$. Moreover, f_n converges to f in $\left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_{p'}^{-\alpha, \infty}(\mathbb{R}^n, \omega) \right)'$;
- (D) for $\max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < p \leq 1, \max \left\{ \frac{N}{N+1}, \frac{N}{N+\alpha+1} \right\} < q \leq 1, \overline{L^2(\mathbb{R}^n, \omega) \cap \dot{F}_p^{\alpha, q}(\mathbb{R}^n, \omega)}$ is the collection of all distributions $f \in \left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_\infty^{-\alpha+N(\frac{1}{p}-1), \infty}(\mathbb{R}^n, \omega) \right)'$ such that there exists a sequence $\{f_n\}_{n=1}^\infty$ in $L^2(\mathbb{R}^n, \omega)$ with $\|f_n - f_m\|_{\dot{F}_p^{\alpha, q}} \rightarrow 0$ as $n, m \rightarrow \infty$. Moreover, f_n converges to f in $\left(L^2(\mathbb{R}^n, \omega) \cap \dot{F}_\infty^{-\alpha+N(\frac{1}{p}-1), \infty}(\mathbb{R}^n, \omega) \right)'$.

The paper is organized as follows. In the next section, we prove Theorem 2.2, which is the Calderón reproducing formula in $L^2 \cap \dot{F}_p^{\alpha, q}$. The main tools are orthogonal estimates in the Dunkl setting. In Sect. 4, we demonstrate the Theorem 2.6, the duality estimates which lead a way for developing the Dunkl–Triebel–Lizorkin space theory. In the last section, we define the Triebel–Lizorkin space in the Dunkl setting in Definition 2.8 and show the spaces is complete by Theorem 2.9.

3 Calderón reproducing formula in $L^2(\mathbb{R}^n, \omega) \cap \dot{F}_p^{\alpha, q}(\mathbb{R}^n, \omega)$

As mentioned before, the Dunkl–Calderón–Zygmund operator theory plays a crucial role. To prove Theorem 2.2, we recall the Dunkl–Calderón–Zygmund singular integral operator and almost orthogonality estimates in the Dunkl setting. See [18] for more details.

Let $C_0^\eta(\mathbb{R}^n)$, $\eta > 0$, denote the space of continuous functions f with compact support and

$$\|f\|_\eta := \sup_{x \neq y} \frac{|f(x) - f(y)|}{\|x - y\|^\eta} < \infty.$$

Definition 3.1 An operator $\mathbf{T} : C_0^\eta(\mathbb{R}^n) \rightarrow (C_0^\eta(\mathbb{R}^n))'$, with $\eta > 0$, is said to be a Dunkl–Calderón–Zygmund singular integral operator with kernel $K(x, y)$ satisfying the following estimates: for some $0 < \varepsilon \leq 1$,

$$|K(x, y)| \leq \frac{C}{\omega(B(x, d(x, y)))} \left(\frac{d(x, y)}{\|x - y\|} \right)^\varepsilon \quad (3.1)$$

for all $x \neq y$;

$$|K(x, y) - K(x', y)| \leq \left(\frac{\|x - x'\|}{\|x - y\|} \right)^\varepsilon \frac{C}{\omega(B(x, d(x, y)))} \quad (3.2)$$

for $\|x - x'\| \leq d(x, y)/2$;

$$|K(x, y) - K(x, y')| \leq \left(\frac{\|y - y'\|}{\|x - y\|} \right)^\varepsilon \frac{C}{\omega(B(x, d(x, y)))} \quad (3.3)$$

for $\|y - y'\| \leq d(x, y)/2$.

A Dunkl–Calderón–Zygmund singular integral operator is said to be the Dunkl–Calderón–Zygmund operator if it extends a bounded operator on $L^2(\mathbb{R}^n)$. Suppose that $\mathbf{T}$ is the Dunkl–Calderón–Zygmund operator. We denote

$$\|\mathbf{T}\|_{dcz} = \|\mathbf{T}\|_{2,2} + \|K\|_{dcz}$$

as the Dunkl–Calderón–Zygmund operator norm, where $\|K\|_{dcz}$ is the minimum of the constants in (3.1)–(3.3).

The following almost orthogonality estimates are important tools in the proof. See [18] for more details. Let $\{\mathbf{S}_k\}_{k \in \mathbb{Z}}$ be approximations to the identity and set $\mathbf{D}_k := \mathbf{S}_k - \mathbf{S}_{k-1}$, then

Lemma 3.2 [18] For $k, j \in \mathbb{Z}$, $\varepsilon > 0$, $\gamma, \varepsilon' \in (0, \varepsilon)$,

$$\left| (D_k D_j)(x, y) \right| \leq C 2^{-|k-j|\varepsilon'} \frac{1}{V(x, y, 2^{-k \vee -j} + d(x, y))} \left(\frac{2^{-k \vee -j}}{2^{-k \vee -j} + d(x, y)} \right)^\gamma,$$

where $a \wedge b = \min \{a, b\}$, $a \vee b = \max \{a, b\}$,
 $V(x, y, r) := \max \{\omega(B(x, r)), \omega(B(y, r))\}$.

Lemma 3.3 [18] Let $\mathbf{T}$ be a Dunkl–Calderón–Zygmund singular integral operator satisfying $\mathbf{T}(1) = \mathbf{T}^*(1) = 0$ and $\mathbf{T}$ is bounded in $L^2(\mathbb{R}^n, \omega)$. Then

$$\left| \int_{\mathbb{R}^n} \int_{\mathbb{R}^n} D_k(x, u) K(u, v) D_j(v, y) d\omega(u) d\omega(v) \right| \leq C 2^{-|k-j|\varepsilon'} \|\mathbf{T}\|_{dcz} \frac{1}{V(x, y, 2^{-k\nu-j} + d(x, y))} \left(\frac{2^{-k\nu-j}}{2^{-k\nu-j} + d(x, y)} \right)^\gamma,$$

where $\gamma, \varepsilon' \in (0, \varepsilon)$, ε is the regularity exponent of the kernel of $\mathbf{T}$ given in (3.2) and (3.3).

Proof of Theorem 2.2: First, as mentioned before, we decompose the identity operator on $L^2(\mathbb{R}^n, \omega)$ by $\mathbf{I} = \mathbf{T}_M + \mathbf{R}_1 + \mathbf{R}_2$. To prove that $\mathbf{T}_M$ is invertible and $(\mathbf{T}_M)^{-1}$, the inverse of $\mathbf{T}_M$, is bounded on $L^2 \cap \dot{F}_p^{\alpha, q}$, we need to estimate $\mathbf{R}_1$ and $\mathbf{R}_2$ on $L^2 \cap \dot{F}_p^{\alpha, q}$ and show that the norm of $\mathbf{R}_1$ and $\mathbf{R}_2$ on $L^2 \cap \dot{F}_p^{\alpha, q}$ are less than 1. To this end, we consider the Dunkl setting, $(\mathbb{R}^n, \|\cdot\|, \omega)$, as a space of homogeneous type in the sense of Coifman and Weiss. The discrete Calderón reproducing formula in the space of homogeneous type is given by the following (see [11])

Theorem 3.4 [11] *Let $\{\mathbf{S}_k\}_{k \in \mathbb{Z}}$ be a Coifman’s approximations to the identity and set $\mathbf{E}_k := \mathbf{S}_k - \mathbf{S}_{k-1}$. Then there exists a family of operators $\{\tilde{\mathbf{E}}_k\}_{k \in \mathbb{Z}}$ such that for any fixed $x_Q \in Q$ with $k \in \mathbb{Z}$ and $Q \in Q_{cw}^k$ are dyadic cubes with the side length 2^{-k-M_0} ,*

$$f(x) = \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} \omega(Q) \tilde{\mathbf{E}}_k(x, x_Q) \mathbf{E}_k(f)(x_Q), \quad (3.4)$$

where the series converge in $L^p(\mathbb{R}^n, \omega)$, $\mathcal{M}(\beta, \gamma, r, x_0)$ and $(\mathcal{M}(\beta, \gamma, r, x_0))'$, the dual space of $\mathcal{M}(\beta, \gamma, r, x_0)$.

Recall the Littlewood–Paley theory and the Triebel–Lizorkin spaces on space of homogeneous type $(\mathbb{R}^n, \|\cdot\|, \omega)$ in the sense of Coifman and Weiss. The discrete Calderón reproducing formula in Theorem 3.4 leads the following discrete q -function on space of homogeneous type $(\mathbb{R}^n, \|\cdot\|, \omega)$:

Definition 3.5 Suppose that $f \in (\mathcal{M}(\beta, \gamma, r, x_0))'$, $\alpha \in \mathbb{R}$, define the Littlewood–Paley q -function $\mathbf{S}_{q, cw}^\alpha(f)$ for the space of homogeneous type $(\mathbb{R}^n, \|\cdot\|, \omega)$ as

$$\mathbf{S}_{q, cw}^\alpha(f)(x) = \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} \left(2^{k\alpha} |\mathbf{E}_k(f)(x_Q)| \right)^q \chi_Q(x) \right\}^{\frac{1}{q}} \quad \text{for } 0 < q < \infty,$$

where $\mathbf{E}_k, Q_{cw}^k$ are given in Theorem 3.4.

Remark 3.6 It is well known that $\mathbb{R}^n$ together with the Euclidean metric and the Dunkl measure is space of homogeneous type in the sense of Coifman and Weiss. See [11] for more details.

Now we return to the estimates of $\mathbf{R}_1$ and $\mathbf{R}_2$. Note that $\mathbf{R}_1$ and $\mathbf{R}_2$ are Dunkl–Calderón–Zygmund operators with $\|\mathbf{R}_1\|_{dcz} + \|\mathbf{R}_2\|_{dcz} \leq C2^{-M\delta}$, $\delta > 0$. The boundedness of $\mathbf{R}_1$ and $\mathbf{R}_2$ on L^2 follows from the Cotlar–Stein Lemma. Moreover, $\|\mathbf{R}_1 + \mathbf{R}_2\|_{2,2} < 1$. See [18] for more details.

To estimate $\mathbf{R}_1$ and $\mathbf{R}_2$ on $\dot{F}_p^{\alpha,q}(\mathbb{R}^n, \omega)$, we establish the following estimates

$$\|\mathbf{S}_q^\alpha(\mathbf{R}_1(f))\|_p \leq C\|\mathbf{S}_{q,cw}^\alpha(\mathbf{R}_1(f))\|_p \leq C\|\mathbf{R}_1\|_{dcz}\|\mathbf{S}_{q,cw}^\alpha(f)\|_p \leq C\|\mathbf{R}_1\|_{dcz}\|\mathbf{S}_q^\alpha(f)\|_p$$

and the similar estimates also hold for $\mathbf{R}_2$. Now we show the above estimates for $\mathbf{R}_1$ by the following steps:

Step 1: $\|\mathbf{S}_q^\alpha(\mathbf{R}_1(f))\|_p \leq C\|\mathbf{S}_{q,cw}^\alpha(\mathbf{R}_1(f))\|_p$.

Indeed we only need to show that for each $f \in L^2(\mathbb{R}^n, \omega)$,

$$\|\mathbf{S}_q^\alpha(f)\|_p \leq C\|\mathbf{S}_{q,cw}^\alpha(f)\|_p$$

since $\mathbf{R}_1$ is bounded on L^2 . By the discrete Calderón reproducing formula of $f \in L^2$ given in Theorem 3.4 we have

$$\begin{aligned} \mathbf{S}_q^\alpha(f)(x) &= \left\{ \sum_{k' \in \mathbb{Z}} \sum_{Q' \in Q_d^{k'}} \left(2^{k'\alpha} \left| \mathbf{D}_{k'} \left(\sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} \omega(Q) \tilde{E}_k(\cdot, x_Q) \mathbf{E}_k(f)(x_Q) \right) (x_{Q'}) \right| \right)^q \chi_{Q'}(x) \right\}^{\frac{1}{q}} \\ &= \left\{ \sum_{k' \in \mathbb{Z}} \sum_{Q' \in Q_d^{k'}} \left(\left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} 2^{k'\alpha} \omega(Q) (D_{k'} \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right| \right)^q \chi_{Q'}(x) \right\}^{\frac{1}{q}}. \end{aligned}$$

By the almost orthogonal estimate given in the Lemma 3.2, we choose $|\alpha| < \varepsilon < 1$ such that

$$\begin{aligned} & \left| (D_{k'} \tilde{E}_k)(x_{Q'}, x_Q) \right| \chi_{Q'}(x) \\ & \leq C2^{-|k'-k|\varepsilon} \frac{1}{V(x_{Q'}, x_Q, 2^{-k' \vee -k} + d(x_{Q'}, x_Q))} \left(\frac{2^{-k' \vee -k}}{2^{-k' \vee -k} + d(x_{Q'}, x_Q)} \right)^\varepsilon \chi_{Q'}(x) \\ & \leq C2^{-|k'-k|\varepsilon} \frac{1}{\omega(B(x_Q, 2^{-k' \vee -k} + d(x, x_Q)))} \left(\frac{2^{-k' \vee -k}}{2^{-k' \vee -k} + d(x, x_Q)} \right)^\varepsilon \chi_{Q'}(x). \end{aligned}$$

Since $d(x, y) = \min_{\sigma \in G} \|\sigma(x) - y\|$, then

$$\begin{aligned}
 & \left| (D_{k'} \tilde{E}_k)(x_{Q'}, x_Q) \right| \chi_{Q'}(x) \\
 & \leq C \sum_{\sigma \in G} 2^{-|k'-k|\varepsilon} \frac{1}{\omega(B(x_Q, 2^{-k'v-k} + \|\sigma(x) - x_Q\|))} \\
 & \quad \times \left(\frac{2^{-k'v-k}}{2^{-k'v-k} + \|\sigma(x) - x_Q\|} \right)^\varepsilon \chi_{Q'}(x) \\
 & \leq C \sum_{\sigma \in G} 2^{-|k'-k|\varepsilon} \frac{1}{\omega(B(\sigma(x), 2^{-k'v-k} + \|\sigma(x) - x_Q\|))} \\
 & \quad \times \left(\frac{2^{-k'v-k}}{2^{-k'v-k} + \|\sigma(x) - x_Q\|} \right)^\varepsilon \chi_{Q'}(x).
 \end{aligned}$$

Let θ satisfies that $\max \left\{ \frac{N}{N+\varepsilon}, \frac{N}{N+\alpha+\varepsilon} \right\} < \theta < \min\{p, q, 1\}$, then

$$\begin{aligned}
 & \left| \sum_{Q \in Q_{cw}^k} 2^{k'\alpha} \omega(Q) (D_{k'} \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right| \chi_{Q'}(x) \\
 & \leq C \sum_{Q \in Q_{cw}^k} \sum_{\sigma \in G} 2^{k'\alpha} 2^{-|k'-k|\varepsilon} \omega(Q) \frac{1}{\omega(B(\sigma(x), 2^{-k'v-k} + \|\sigma(x) - x_Q\|))} \\
 & \quad \times \left(\frac{2^{-k'v-k}}{2^{-k'v-k} + \|\sigma(x) - x_Q\|} \right)^\varepsilon \left| \mathbf{E}_k f(x_Q) \right| \chi_{Q'}(x) \\
 & \leq C \sum_{\sigma \in G} \left\{ \sum_{Q \in Q_{cw}^k} 2^{-|k'-k|\varepsilon\theta} 2^{(k'-k)\alpha\theta} \omega(Q)^\theta \frac{1}{\omega(B(\sigma(x), 2^{-k'v-k} + \|\sigma(x) - x_Q\|))^\theta} \right. \\
 & \quad \times \left. \left(\frac{2^{-k'v-k}}{2^{-k'v-k} + \|\sigma(x) - x_Q\|} \right)^{\theta\varepsilon} \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \right\}^{\frac{1}{\theta}} \chi_{Q'}(x).
 \end{aligned}$$

Denote c_Q as the center of Q . Set

$$A_0 = \{Q \in Q_{cw}^k : \|c_Q - \sigma(x)\| \leq 2^{-kv-k'}\},$$

and

$$A_\ell = \{Q \in Q_{cw}^k : 2^{\ell-1+(-kv-k')} < \|c_Q - \sigma(x)\| \leq 2^{\ell+(-kv-k')}\}$$

for $\ell \in \mathbb{N}$.

For $Q \in Q_{cw}^k$, we have

$$\omega(Q) \chi_Q(z) \sim \omega(B(z, 2^{-k-M_0})) \chi_Q(z) \sim \omega(B(\sigma(z), 2^{-k-M_0})) \chi_Q(z), \quad \text{for } \sigma \in G,$$

and

$$\omega(B(\sigma(z), 2^{-kv-k'})) \leq C 2^{((-kv-k')-(-k-M_0))N} \omega(B(\sigma(z), 2^{-k-M_0})).$$

Hence

$$\begin{aligned}
& \sum_{Q \in Q_{cw}^k} 2^{-|k'-k|\varepsilon\theta} 2^{(k'-k)\alpha\theta} \omega(Q)^\theta \frac{1}{\omega(B(\sigma(x), 2^{-k'\vee-k} + \|\sigma(x) - x_Q\|))^\theta} \\
& \quad \times \left(\frac{2^{-k'\vee-k}}{2^{-k'\vee-k} + \|\sigma(x) - x_Q\|} \right)^{\theta\varepsilon} \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \\
& = \sum_{\ell=0}^{\infty} \sum_{Q \in A_{\ell}} 2^{-|k'-k|\varepsilon\theta} 2^{(k'-k)\alpha\theta} \frac{\omega(Q)^{\theta-1}}{\omega(B(\sigma(x), 2^{-k'\vee-k} + \|\sigma(x) - x_Q\|))^\theta} \\
& \quad \times \left(\frac{2^{-k'\vee-k}}{2^{-k'\vee-k} + \|\sigma(x) - x_Q\|} \right)^{\theta\varepsilon} \frac{1}{\omega(B(\sigma(x), 2^{-k'\vee-k} + \|\sigma(x) - x_Q\|))^\theta} \\
& \quad \times \omega(Q) \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \\
& \leq C 2^{-|k'-k|\varepsilon\theta} 2^{(k'-k)\alpha\theta} 2^{(-k-M_0-(-k'\vee-k))N(\theta-1)} \\
& \quad \times \sum_{\ell=0}^{\infty} \frac{\omega(B(\sigma(x), 2^{-k'\vee-k}))^{\theta-1}}{\omega(B(\sigma(x), 2^{\ell-1+(-k'\vee-k)}))^{\theta-1}} \left(\frac{1}{2^{\ell-1}} \right)^{\theta\varepsilon} \frac{1}{\omega(B(\sigma(x), 2^{\ell-1+(-k'\vee-k)}))^\theta} \\
& \quad \times \int_{\|\sigma(x)-z\| \leq 2 \times 2^{\ell+(-k'\vee-k)}} \sum_{Q \in A_{\ell}} \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \chi_Q(z) dz \\
& \leq C 2^{-M_0N(\theta-1)} 2^{-|k'-k|\varepsilon\theta} 2^{(-k-(-k'\vee-k))N(\theta-1)} 2^{(k'-k)\alpha\theta} \\
& \quad \times \sum_{\ell=0}^{\infty} \frac{1}{2^{(\ell-1)(N(\theta-1)+\theta\varepsilon)}} \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \chi_Q \right) (\sigma(\cdot)) \\
& \leq C 2^{-M_0N(\theta-1)} 2^{-|k'-k|\varepsilon\theta} 2^{(-k-(-k'\vee-k))N(\theta-1)+(k'-k)\alpha\theta} \\
& \quad \times \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \chi_Q \right) (\sigma(\cdot)),
\end{aligned}$$

where $\mathbf{M}$ denote the Hardy–Littlewood maximal operator on $(\mathbb{R}^n, \|\cdot\|, \omega)$. Therefore,

$$\begin{aligned}
& \left| \sum_{Q \in Q_{cw}^k} 2^{k'\alpha} \omega(Q) (D_{k'} \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right| \chi_{Q'}(x) \\
& \leq C \sum_{\sigma \in G} 2^{-M_0N(1-\frac{1}{\theta})} 2^{-|k'-k|\varepsilon\theta} 2^{(-k-(-k'\vee-k))N(1-\frac{1}{\theta})+(k'-k)\alpha} \\
& \quad \times \left\{ \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \chi_Q \right) (\sigma(\cdot)) \right\}^{\frac{1}{\theta}} \chi_{Q'}(x).
\end{aligned}$$

For $|\alpha| < \varepsilon$, $\max \left\{ \frac{N}{N+\varepsilon}, \frac{N}{N+\alpha+\varepsilon} \right\} < \theta < \min \{p, q, 1\}$, it is obvious that

$$\sup_{k' \in \mathbb{Z}} \sum_{k \in \mathbb{Z}} 2^{-|k-k'| \varepsilon} 2^{(-k-(-k' \vee -k))N(1-\frac{1}{\theta})+(k'-k)\alpha} < \infty.$$

For $1 < q < \infty$, by Hölder's inequality, we have

$$\begin{aligned} & \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} 2^{k'\alpha} \omega(Q) (D_{k'} \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right|_{\chi_{Q'}(x)}^q \\ & \leq C \left\{ \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} 2^{-M_0 N(1-\frac{1}{\theta})} 2^{-|k-k'| \varepsilon} 2^{(-k-(-k' \vee -k))N(1-\frac{1}{\theta})+(k'-k)\alpha} \right. \\ & \quad \times \left. \left\{ \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \chi_Q \right) (\sigma(\cdot)) \right\}^{\frac{1}{\theta}} \right\}^q \chi_{Q'}(x) \\ & \leq C 2^{-M_0 N(1-\frac{1}{\theta})q} \left\{ \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} 2^{-|k-k'| \varepsilon} 2^{(-k-(-k' \vee -k))N(1-\frac{1}{\theta})+(k'-k)\alpha} \right\}^{\frac{q}{q'}} \\ & \quad \times \left\{ \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} 2^{-|k-k'| \varepsilon} 2^{(-k-(-k' \vee -k))N(1-\frac{1}{\theta})+(k'-k)\alpha} \right. \\ & \quad \times \left. \left\{ \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \chi_Q \right) (\sigma(\cdot)) \right\}^{\frac{q}{\theta}} \right\} \chi_{Q'}(x) \\ & \leq C 2^{-M_0 N(1-\frac{1}{\theta})q} \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} 2^{-|k-k'| \varepsilon} 2^{(-k-(-k' \vee -k))N(1-\frac{1}{\theta})+(k'-k)\alpha} \\ & \quad \times \left\{ \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \chi_Q \right) (\sigma(\cdot)) \right\}^{\frac{q}{\theta}} \chi_{Q'}(x). \end{aligned}$$

For $\max \left\{ \frac{N}{N+\varepsilon}, \frac{N}{N+\alpha+\varepsilon} \right\} < q \leq 1$, by q -inequality we have

$$\begin{aligned}
& \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} 2^{k'\alpha} (D_{k'} \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right|^q \chi_{Q'}(x) \\
& \leq C 2^{-M_0 N(1-\frac{1}{\theta})q} \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} 2^{-|k-k'|\varepsilon q} 2^{(-k-(-k'\vee-k))N(1-\frac{1}{\theta})q+(k'-k)\alpha q} \\
& \quad \times \left\{ \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \chi_Q \right) (\sigma(\cdot)) \right\}^{\frac{q}{\theta}} \chi_{Q'}(x).
\end{aligned}$$

For $|\alpha| < \varepsilon$, $\max \left\{ \frac{N}{N+\varepsilon}, \frac{N}{N+\alpha+\varepsilon} \right\} < \theta < \min \{p, q, 1\}$, it is obvious that

$$\sup_k \sum_{k' \in \mathbb{Z}} 2^{-|k-k'|\varepsilon} 2^{(-k-(-k'\vee-k))N(1-\frac{1}{\theta})+(k'-k)\alpha} < \infty,$$

then

$$\begin{aligned}
& \left\{ \sum_{k' \in \mathbb{Z}} \sum_{Q' \in Q_d^{k'}} \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} 2^{k'\alpha} \omega(Q) (D_{k'} \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right|^q \chi_{Q'}(x) \right\}^{\frac{1}{q}} \\
& \leq C 2^{-M_0 N(1-\frac{1}{\theta})} \left\{ \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} \left\{ \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} \left| 2^{k\alpha} \mathbf{E}_k f(x_Q) \right|^\theta \chi_Q \right) (\sigma(\cdot)) \right\}^{\frac{q}{\theta}} \right\}^{\frac{1}{q}}.
\end{aligned}$$

By using Fefferman–Stein vectored maximal function inequality (see [8]) with $\theta < \min \{p, q, 1\}$, we have

$$\begin{aligned}
& \left\| \left\{ \sum_{k' \in \mathbb{Z}} \sum_{Q' \in Q_d^{k'}} \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} 2^{k'\alpha} \omega(Q) (D_{k'} \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right|^q \chi_{Q'}(\cdot) \right\}^{\frac{1}{q}} \right\|_p \\
& \leq C 2^{-M_0 N(1-\frac{1}{\theta})} \sum_{\sigma \in G} \left\| \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} \left(2^{k\alpha} \left| \mathbf{E}_k(f)(x_Q) \right| \right)^q \chi_Q(\sigma(\cdot)) \right\}^{\frac{1}{q}} \right\|_p \\
& \leq C 2^{-M_0 N(1-\frac{1}{\theta})} \left\| \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} \left(2^{k\alpha} \left| \mathbf{E}_k(f)(x_Q) \right| \right)^q \chi_Q(\cdot) \right\}^{\frac{1}{q}} \right\|_p,
\end{aligned}$$

where the last inequality follows from the fact that G is a finite group and

$$\int_{\mathbb{R}^n} f(\sigma(x)) d\omega(x) = \int_{\mathbb{R}^n} f(x) d\omega(x).$$

Thus,

$$\|\mathbf{S}_q^\alpha(f)\|_p \leq C 2^{-M_0 N(1-\frac{1}{\theta})} \|\mathbf{S}_{q,cw}^\alpha(f)\|_p.$$

Step 2: $\|\mathbf{S}_{q,cw}^\alpha(\mathbf{R}_1(f))\|_p \leq C \|\mathbf{R}_1\|_{dcz} \|\mathbf{S}_{q,cw}^\alpha(\mathbf{R}_1(f))\|_p.$

The L^2 boundedness of $\mathbf{R}_1$ together with the discrete Calderón reproducing formula of $f \in L^2$ on space of homogeneous type given in (3.4) yields

$$\begin{aligned} & \|\mathbf{S}_{q,cw}^\alpha(\mathbf{R}_1(f))\|_p \\ &= \left\| \left\{ \sum_{k' \in \mathbb{Z}} \sum_{Q' \in Q_{cw}^{k'}} \left(2^{k'\alpha} \left| \mathbf{E}_{k'}(\mathbf{R}_1(f))(x_{Q'}) \right| \right)^q \chi_{Q'}(\cdot) \right\}^{\frac{1}{q}} \right\|_p \\ &= \left\| \left\{ \sum_{k' \in \mathbb{Z}} \sum_{Q' \in Q_{cw}^{k'}} \left(2^{k'\alpha} \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} \omega(Q)(E_{k'} R_1 \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right| \right)^q \chi_{Q'}(\cdot) \right\}^{\frac{1}{q}} \right\|_p. \end{aligned}$$

Applying the almost orthogonal estimate given in Lemma 3.3 to $(E_{k'} R_1 \tilde{E}_k)(x_{Q'}, x_Q)$, we obtain that for $|\alpha| < \varepsilon < 1$,

$$\begin{aligned} & \left| (E_{k'} R_1 \tilde{E}_k)(x_{Q'}, x_Q) \right| \chi_{Q'}(x) \\ & \leq C \|\mathbf{R}_1\|_{dcz} 2^{-|k'-k|\varepsilon} \frac{1}{V(x_{Q'}, x_Q, 2^{-k'\vee-k} + d(x_{Q'}, x_Q))} \\ & \quad \times \left(\frac{2^{-k'\vee-k}}{2^{-k'\vee-k} + d(x_{Q'}, x_Q)} \right)^\varepsilon \chi_{Q'}(x) \\ & \leq C \|\mathbf{R}_1\|_{dcz} \sum_{\sigma \in G} 2^{-|k'-k|\varepsilon} \frac{1}{V(x_{Q'}, x_Q, 2^{-k'\vee-k} + \|\sigma(x_{Q'}) - x_Q\|)} \\ & \quad \times \left(\frac{2^{-k'\vee-k}}{2^{-k'\vee-k} + \|\sigma(x_{Q'}) - x_Q\|} \right)^\varepsilon \chi_{Q'}(x) \\ & \leq C \|\mathbf{R}_1\|_{dcz} \sum_{\sigma \in G} 2^{-|k'-k|\varepsilon} \frac{1}{\omega(B(\sigma(x), 2^{-k'\vee-k} + \|\sigma(x) - x_Q\|))} \\ & \quad \times \left(\frac{2^{-k'\vee-k}}{2^{-k'\vee-k} + \|\sigma(x) - x_Q\|} \right)^\varepsilon \chi_{Q'}(x) \end{aligned}$$

where we use the fact that if $x \in Q'$, then $2^{-k\nu-k'} + \|\sigma(x_{Q'}) - x_Q\| \sim 2^{-k\nu-k'} + \|\sigma(x) - x_Q\|$ and $\omega(B(x_{Q'}, 2^{-k\nu-k'} + \|\sigma(x) - x_Q\|))\chi_{Q'}(x) \sim \omega(B(\sigma(x), 2^{-k\nu-k'} + \|\sigma(x) - x_Q\|))\chi_{Q'}(x)$.

Similar to the estimate in **Step 1**, for $\max\left\{\frac{N}{N+\varepsilon}, \frac{N}{N+\varepsilon+\alpha}\right\} < \theta < \min\{p, q, 1\}$ we have

$$\begin{aligned} & \left| \sum_{Q \in Q_{cw}^k} 2^{k'\alpha} \omega(Q)(E_{k'} R_1 \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right| \chi_{Q'}(x) \\ & \leq C \|\mathbf{R}_1\|_{dcz} \sum_{\sigma \in G} 2^{-M_0 N(1-\frac{1}{\theta})} 2^{-|k'-k|\varepsilon} 2^{(-k-(-k\nu-k'))N(1-\frac{1}{\theta})+(k'-k)\alpha} \\ & \quad \times \left\{ \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} \left| 2^{k\alpha} \mathbf{E}_k(f)(x_Q) \right|^\theta \chi_Q \right) (\sigma(\cdot)) \right\}^{\frac{1}{\theta}} \chi_{Q'}(x). \end{aligned}$$

For $1 < q < \infty$, by Hölder's inequality, we have

$$\begin{aligned} & \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} 2^{k'\alpha} \omega(Q)(E_{k'} R_1 \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right|^q \chi_{Q'}(x) \\ & \leq C 2^{-M_0 N(1-\frac{1}{\theta})q} \|\mathbf{R}_1\|_{dcz}^q \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} 2^{-|k-k'|\varepsilon} 2^{(-k-(-k\nu-k'))N(1-\frac{1}{\theta})+(k'-k)\alpha} \\ & \quad \times \left\{ \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} \left| 2^{k\alpha} \mathbf{E}_k(f)(x_Q) \right|^\theta \chi_Q \right) (\sigma(\cdot)) \right\}^{\frac{q}{\theta}} \chi_{Q'}(x). \end{aligned} \quad (3.5)$$

For $\max\left\{\frac{N}{N+\varepsilon}, \frac{N}{N+\varepsilon+\alpha}\right\} < q \leq 1$, by q -inequality we have

$$\begin{aligned} & \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} 2^{k'\alpha} \omega(Q)(E_{k'} R_1 \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right|^q \chi_{Q'}(x) \\ & \leq C 2^{-M_0 N(1-\frac{1}{\theta})q} \|\mathbf{R}_1\|_{dcz}^q \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} 2^{-|k-k'|\varepsilon q} 2^{(-k-(-k\nu-k'))N(1-\frac{1}{\theta})q+(k'-k)\alpha q} \\ & \quad \times \left\{ \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} \left| 2^{k\alpha} \mathbf{E}_k(f)(x_Q) \right|^\theta \chi_Q \right) (\sigma(\cdot)) \right\}^{\frac{q}{\theta}} \chi_{Q'}(x). \end{aligned} \quad (3.6)$$

Then (3.5)–(3.6) implies that

$$\left\{ \sum_{k' \in \mathbb{Z}} \sum_{Q' \in Q_{cw}^{k'}} \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} 2^{k'\alpha} \omega(Q) (E_{k'} R_1 \tilde{E}_k)(x_{Q'}, x_Q) \mathbf{E}_k(f)(x_Q) \right|^q \chi_{Q'}(x) \right\}^{\frac{1}{q}}$$

$$\leq C 2^{-M_0 N(1-\frac{1}{\theta})} \|\mathbf{R}_1\|_{dcz} \left\{ \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} \left\{ \mathbf{M} \left(\sum_{Q \in Q_{cw}^k} |2^{k\alpha} \mathbf{E}_k(f)(x_Q)|^\theta \chi_Q \right) (\sigma(\cdot)) \right\}^{\frac{q}{\theta}} \right\}^{\frac{1}{q}}.$$

The Fefferman–Stein vector valued maximal function inequality with $\theta < \min \{p, q, 1\}$ yields

$$\|\mathbf{S}_{q,cw}^\alpha(\mathbf{R}_1(f))\|_p$$

$$\leq C 2^{-M_0 N(1-\frac{1}{\theta})} \|\mathbf{R}_1\|_{dcz} \left\| \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_{cw}^k} |2^{k\alpha} \mathbf{E}_k(f)(x_Q)|^q \chi_Q(\sigma(\cdot)) \right\}^{\frac{1}{q}} \right\|_p$$

$$\leq C 2^{-M_0 N(1-\frac{1}{\theta})} \|\mathbf{R}_1\|_{dcz} \|\mathbf{S}_{q,cw}^\alpha(f)\|_p.$$

Applying the similar proof, we also have

$$\|\mathbf{S}_{q,cw}^\alpha(\mathbf{R}_2(f))\|_p \leq C 2^{-M_0 N(1-\frac{1}{\theta})} \|\mathbf{R}_2\|_{dcz} \|\mathbf{S}_{q,cw}^\alpha(f)\|_p.$$

Step 3: $\|\mathbf{S}_{q,cw}^\alpha(f)\|_p \leq C \|\mathbf{S}_q^\alpha(f)\|_p.$

To show this estimate, the key point is to write

$$f(x) = \mathbf{T}_M(f)(x) + \mathbf{R}_1(f)(x) + \mathbf{R}_2(f)(x).$$

Recall the estimates in **Step 2** for $\max \left\{ \frac{N}{N+\varepsilon}, \frac{N}{N+\alpha+\varepsilon} \right\} \leq p < \infty$,

$$\|\mathbf{S}_{q,cw}^\alpha(\mathbf{R}_1(f))\|_p \leq C 2^{-M_0 N(1-\frac{1}{\theta})} \|\mathbf{R}_1\|_{dcz} \|\mathbf{S}_{q,cw}^\alpha(f)\|_p,$$

and

$$\|\mathbf{S}_{q,cw}^\alpha(\mathbf{R}_2(f))\|_p \leq C 2^{-M_0 N(1-\frac{1}{\theta})} \|\mathbf{R}_2\|_{dcz} \|\mathbf{S}_{q,cw}^\alpha(f)\|_p.$$

Since $\|\mathbf{R}_1\|_{dcz} + \|\mathbf{R}_2\|_{dcz} \leq C 2^{-M\delta}$, $\delta > 0$, and we choose M is sufficiently larger than M_0 , then we can set $C 2^{-M_0 N(1-\frac{1}{\theta})} (\|\mathbf{R}_1\|_{dcz} + \|\mathbf{R}_2\|_{dcz}) \leq \min \left\{ \left(\frac{1}{2} \right)^{\frac{1}{p}}, \frac{1}{2} \right\}$, then for $\max \left\{ \frac{N}{N+\varepsilon}, \frac{N}{N+\alpha+\varepsilon} \right\} < p \leq 1$, by using p -inequality, we have

$$\|\mathbf{S}_{q,cw}^\alpha(f)\|_p^p \leq \|\mathbf{S}_{q,cw}^\alpha(\mathbf{T}_M(f) + \mathbf{R}_1(f) + \mathbf{R}_2(f))\|_p^p \leq \|\mathbf{S}_{q,cw}^\alpha(\mathbf{T}_M(f))\|_p^p + \frac{1}{2}\|\mathbf{S}_{q,cw}^\alpha(f)\|_p^p,$$

and for $1 < p < \infty$, by using Minkowski's inequality we have

$$\begin{aligned}\|\mathbf{S}_{q,cw}^\alpha(f)\|_p &\leq \|\mathbf{S}_{q,cw}^\alpha(\mathbf{T}_M(f))\|_p + \|\mathbf{S}_{q,cw}^\alpha(\mathbf{R}_1(f))\|_p + \|\mathbf{S}_{q,cw}^\alpha(\mathbf{R}_2(f))\|_p \\ &\leq C\|\mathbf{S}_{q,cw}^\alpha(\mathbf{T}_M(f))\|_p + \frac{1}{2}\|\mathbf{S}_{q,cw}^\alpha(f)\|_p.\end{aligned}$$

Hence,

$$\|\mathbf{S}_{q,cw}^\alpha(f)\|_p \leq C_p\|\mathbf{S}_{q,cw}^\alpha(\mathbf{T}_M(f))\|_p.$$

Claim: $\|\mathbf{S}_{q,cw}^\alpha(\mathbf{T}_M(f))\|_p \leq C_p\|\mathbf{S}_q^\alpha(f)\|_p$.

Indeed, observing that

$$\mathbf{T}_M(f)(x) = \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(f)(x_Q),$$

and

$$|\mathbf{E}_{k'} \mathbf{T}_M(f)(x)| \leq \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) \left| (E_{k'} D_k^M)(x, x_Q) \right| \left| \mathbf{D}_k(f)(x_Q) \right|.$$

Following the same proof as in **Step 1**, there exists a constant $C > 0$, such that

$$\begin{aligned}&\left| (E_{k'} D_k^M)(x, x_Q) \right| \chi_{Q'}(x) \\ &\leq C \sum_{\sigma \in G} 2^{-|k-k'|\varepsilon} \frac{1}{\omega(B(\sigma(x), 2^{-k\vee-k'} + \|\sigma(x) - x_Q\|))} \\ &\times \left(\frac{2^{-k\vee-k'}}{2^{-k\vee-k'} + \|\sigma(x) - x_Q\|} \right)^\varepsilon \chi_{Q'}(x).\end{aligned}$$

Therefore for $\max\left\{\frac{N}{N+\varepsilon}, \frac{N}{N+\alpha+\varepsilon}\right\} < \theta < \min\{p, q, 1\}$, we have

$$\begin{aligned}&\left| \sum_{Q \in Q_d^k} 2^{k'\alpha} \omega(Q) (E_{k'} D_k^M)(x, x_Q) \mathbf{D}_k(f)(x_Q) \right| \chi_{Q'}(x) \\ &\leq C \sum_{\sigma \in G} 2^{-MN(1-\frac{1}{\theta})} 2^{-|k-k'|\varepsilon} 2^{(-k-(-k'\vee-k))N(1-\frac{1}{\theta})+(k'-k)\alpha} \\ &\times \left\{ \mathbf{M} \left(\sum_{Q \in Q_d^k} \left| 2^{k\alpha} \mathbf{D}_k(f)(x_Q) \right|^\theta \right) (\sigma(\cdot)) \right\}^{\frac{1}{\theta}} \chi_{Q'}(x).\end{aligned}$$

For $1 < q < \infty$, $\max\left\{\frac{N}{N+\varepsilon}, \frac{N}{N+\alpha+\varepsilon}\right\} < \theta < \min\{p, q, 1\}$, by Hölder's inequality, we obtain

$$\begin{aligned}
 & \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k' \alpha} \omega(Q) (E_{k'} D_k^M)(x, x_Q) \mathbf{D}_k(f)(x_Q) \right|^q \chi_{Q'}(x) \\
 & \leq C 2^{-MN(1-\frac{1}{\theta})q} \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} 2^{-|k-k'|\varepsilon} 2^{(-k-(-k' \vee -k))N(1-\frac{1}{\theta})+(k'-k)\alpha} \\
 & \quad \times \left\{ \mathbf{M} \left(\sum_{Q \in Q_d^k} \left| 2^{k\alpha} \mathbf{D}_k(f)(x_Q) \right|^\theta \right) (\sigma(\cdot)) \right\}^{\frac{q}{\theta}} \chi_{Q'}(x).
 \end{aligned} \tag{3.7}$$

For $\max \left\{ \frac{N}{N+\varepsilon}, \frac{N}{N+\alpha+\varepsilon} \right\} < q \leq 1$, by q -inequality, we have

$$\begin{aligned}
 & \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k' \alpha} \omega(Q) (E_{k'} D_k^M)(x, x_Q) \mathbf{D}_k(f)(x_Q) \right|^q \chi_{Q'}(x) \\
 & \leq C 2^{-MN(1-\frac{1}{\theta})q} \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} 2^{-|k-k'|\varepsilon q} 2^{(-k-(-k' \vee -k))N(1-\frac{1}{\theta})q+(k'-k)\alpha q} \\
 & \quad \times \left\{ \mathbf{M} \left(\sum_{Q \in Q_d^k} \left| 2^{k\alpha} \mathbf{D}_k(f)(x_Q) \right|^\theta \right) (\sigma(\cdot)) \right\}^{\frac{q}{\theta}} \chi_{Q'}(x).
 \end{aligned} \tag{3.8}$$

Thus (3.7)–(3.8) implies that

$$\begin{aligned}
 & \left\{ \sum_{k' \in \mathbb{Z}} \sum_{Q' \in Q_{cw}^{k'}} \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k' \alpha} \omega(Q) (E_{k'} D_k^M)(x, x_Q) \mathbf{D}_k(f)(x_Q) \right|^q \chi_{Q'}(x) \right\}^{\frac{1}{q}} \\
 & \leq C 2^{-MN(1-\frac{1}{\theta})} \left\{ \sum_{\sigma \in G} \sum_{k \in \mathbb{Z}} \left\{ \mathbf{M} \left(\sum_{Q \in Q_d^k} \left| 2^{k\alpha} \mathbf{D}_k(f)(x_Q) \right|^\theta \right) (\sigma(\cdot)) \right\}^{\frac{q}{\theta}} \right\}^{\frac{1}{q}}.
 \end{aligned}$$

The Fefferman–Stein vector valued maximal function inequality with $\theta < \min \{p, q, 1\}$ yields

$$\begin{aligned} \|\mathbf{S}_{q,cw}^\alpha(\mathbf{T}_M(f))\|_p &\leq C 2^{-MN(1-\frac{1}{\theta})} \left\| \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{k\alpha} |\mathbf{D}_k(f)(x_Q)| \right)^q \chi_Q(\cdot) \right\}^{\frac{1}{q}} \right\|_p \\ &\leq C \|\mathbf{S}_q^\alpha(f)\|_p, \end{aligned}$$

where M is a fixed constant.

The proof of **Step 3** is complete.

Observing that $f(x) = \mathbf{T}_M(f)(x) + \mathbf{R}_1(f)(x) + \mathbf{R}_2(f)(x)$ and applying the above estimates, we have $\|\mathbf{S}_q^\alpha(\mathbf{R}_1 + \mathbf{R}_2)(f)\|_p \leq C(\|\mathbf{R}_1\|_{dcz} + \|\mathbf{R}_2\|_{dcz})\|\mathbf{S}_q^\alpha(f)\|_p$, so

$$\|\mathbf{S}_q^\alpha(\mathbf{I} - \mathbf{T}_M)(f)\|_p = C\|\mathbf{S}_q^\alpha(\mathbf{R}_1 + \mathbf{R}_2)(f)\|_p \leq \frac{1}{2}\|\mathbf{S}_q^\alpha(f)\|_p.$$

Similarly, for $(\mathbf{T}_M)^{-1}(f)$, we also have $\|\mathbf{S}_q^\alpha(\mathbf{T}_M(f))^{-1}\|_p \leq C\|\mathbf{S}_q^\alpha(f)\|_p$. If there exist a function $h \in L^2$ and set $h = (\mathbf{T}_M)^{-1}f$, we obtain

$$f(x) = \mathbf{T}_M(h)(x) = \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(h)(x_Q), \quad (3.9)$$

where $\|f\|_2 \sim \|h\|_2$, and $\|f\|_{\dot{F}_p^{\alpha,q}} \sim \|h\|_{\dot{F}_p^{\alpha,q}}$, $|\alpha| < 1$, $\max\left\{\frac{N}{N+1}, \frac{N}{N+\alpha+1}\right\} < p < \infty$, $\max\left\{\frac{N}{N+1}, \frac{N}{N+\alpha+1}\right\} < q < \infty$.

It remains to show that the series (3.9) converges in $L^2 \cap \dot{F}_p^{\alpha,q}$. To this end, we only need to prove

$$\left\| \mathbf{S}_q^\alpha \left(\sum_{|k|>m} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(h)(x_Q) \right) \right\|_p \longrightarrow 0, \quad \text{as } m \rightarrow \infty.$$

Repeating the same proof in **Step 1**,

$$\begin{aligned} &\left\| \mathbf{S}_q^\alpha \left(\sum_{|k|>m} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(h)(x_Q) \right) \right\|_p \\ &\leq C \left\| \left\{ \sum_{|k|>m} \sum_{Q \in Q_d^k} \left(2^{k\alpha} |\mathbf{D}_k(h)(x_Q)| \right)^q \chi_Q(x) \right\}^{\frac{1}{q}} \right\|_p, \end{aligned}$$

where by the fact $\|h\|_{\dot{F}_p^{\alpha,q}} \sim \|f\|_{\dot{F}_p^{\alpha,q}}$, the last term tends to 0 as $m \rightarrow \infty$.

The proof of Theorem 2.2 is complete. $\square$

4 Duality estimates

The duality estimates will be a key idea in developing the Dunkl–Triebel–Lizorkin space theory. Now we show Theorem 2.6.

Proof of Theorem 2.6(A) Applying the weak-type discrete Calderón-type reproducing formula given in Theorem 2.2 for $f \in L^2 \cap \dot{F}_p^{\alpha,q}$, we write

$$f(x) = \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(h)(x_Q),$$

where $\|h\|_2 \sim \|f\|_2$, and $\|\mathbf{S}_q^\alpha(h)\|_p \sim \|\mathbf{S}_q^\alpha(f)\|_p$.

For $1 < q < \infty$, by Hölder's inequality

$$\begin{aligned} |\langle f, g \rangle| &= \left| \left\langle \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(\cdot, x_Q) \mathbf{D}_k(h)(x_Q), g(\cdot) \right\rangle \right| \\ &\leq \int_{\mathbb{R}^n} \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k\alpha} |\mathbf{D}_k(h)(x_Q)| 2^{-k\alpha} |\mathbf{D}_k^M(g)(x_Q)| \chi_Q(x) d\omega(x) \\ &\leq C \int_{\mathbb{R}^n} \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{k\alpha} |\mathbf{D}_k(h)(x_Q)| \right)^q \chi_Q(x) \right\}^{\frac{1}{q}} \\ &\quad \times \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{-k\alpha} |\mathbf{D}_k^M(g)(x_Q)| \right)^{q'} \chi_Q(x) \right\}^{\frac{1}{q'}} d\omega(x) \\ &\leq C \left\{ \int_{\mathbb{R}^n} |\mathbf{S}_q^\alpha(h)(x)|^p d\omega(x) \right\}^{\frac{1}{p}} \left\{ \int_{\mathbb{R}^n} |\mathbf{S}_{q'}^{-\alpha}(g)(x)|^{p'} d\omega(x) \right\}^{\frac{1}{p'}} \\ &\leq C \|f\|_{\dot{F}_p^{\alpha,q}} \|g\|_{\dot{F}_{p'}^{-\alpha,q'}}. \end{aligned}$$

□

Proof of Theorem 2.6(B) Set $\Omega_\ell = \left\{ x \in \mathbb{R}^n : \mathbf{S}_q^\alpha(h)(x) > 2^\ell \right\}$, and

$$B_\ell = \left\{ Q : \omega(Q \cap \Omega_\ell) > \frac{1}{2} \omega(Q) \text{ and } \omega(Q \cap \Omega_{\ell+1}) \leq \frac{1}{2} \omega(Q) \right\},$$

where Q are dyadic cubes. Denote $B_\ell^* := \{Q^*\}$ as the maximal dyadic cubes in B_ℓ for $\ell \in \mathbb{Z}$.

Claim: the Calderón reproducing formula can be rewrite as:

$$f(x) = \sum_{\ell \in \mathbb{Z}} \sum_{Q_\ell^* \in B_\ell^*} \sum_{Q \subset Q_\ell^*} \omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(h)(x_Q).$$

In order to prove the above claim, we only need to show that if the dyadic cube $Q \notin B_\ell$ for all $\ell \in \mathbb{Z}$, then

$$\omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(h)(x_Q) = 0. \quad (4.1)$$

Observe that by the stopping time argument, each dyadic cube Q can be in one and only one B_ℓ , that is, if Q belongs to both B_ℓ and $B_{\ell'}$, then $\ell = \ell'$. We now assume that $\omega(Q) \neq 0$. Otherwise, the equality (4.1) holds obviously. Note that $\omega(\Omega_\ell) < 2^{-2\ell} \|\mathbf{S}_q^\alpha(h)\|_2^2 \rightarrow 0$ as $\ell \rightarrow +\infty$. As a consequence, if $Q \notin B_\ell$ for all $\ell \in \mathbb{Z}$, then $\omega(Q \cap \Omega_\ell) \leq \frac{1}{2}\omega(Q)$ for all $\ell \in \mathbb{Z}$ since, otherwise, there exists an $\ell_0 \in \mathbb{Z}$, such that $\omega(Q \cap \Omega_{\ell_0}) > \frac{1}{2}\omega(Q)$. However, $\omega(Q \cap \Omega_\ell) \rightarrow 0$ as $\ell \rightarrow +\infty$ and $\{\omega(Q \cap \Omega_\ell)\}_\ell$ is a decreasing sequence. So there must be a critical index ℓ_1 such that $\omega(Q \cap \Omega_{\ell_1}) > \frac{1}{2}\omega(Q)$ and $\omega(Q \cap \Omega_{\ell_1+1}) \leq \frac{1}{2}\omega(Q)$, that is $Q \in B_{\ell_1}$. This is contradiction to the fact that Q is not in B_ℓ for all $\ell \in \mathbb{Z}$.

Since $\omega(Q \cap \Omega_\ell) \leq \frac{1}{2}\omega(Q)$ for all $\ell \in \mathbb{Z}$, then $\omega(Q \cap \Omega_\ell^c) \geq \frac{1}{2}\omega(Q)$ for all $\ell \in \mathbb{Z}$. Set $K = \left\{x \in \mathbb{R}^n, \mathbf{S}_q^\alpha(h)(x) = 0\right\}$. Note that

$$\bigcap_{\ell \in \mathbb{Z}} \Omega_\ell^c = \bigcap_{\ell \in \mathbb{Z}} \left\{x \in \mathbb{R}^n : \mathbf{S}_q^\alpha(h)(x) \leq 2^\ell\right\} = K.$$

Thus

$$\omega(Q \cap K) = \lim_{\ell \rightarrow -\infty} \omega(Q \cap \Omega_\ell^c) \geq \frac{1}{2}\omega(Q) > 0$$

for all $x \in K$, $0 = \mathbf{S}_q^\alpha(h)(x) = \left\{\sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{k\alpha} |\mathbf{D}_k(h)(x_Q)|\right)^q \chi_Q(x)\right\}^{\frac{1}{q}}$, then $|\mathbf{D}_k(h)(x_Q)| = 0$. Hence the claim is proved.

Then by Hölder's inequality and p -inequality with $p \leq 1$, we have

$$\begin{aligned}
 & | \langle f, g \rangle | \\
 &= \left| \sum_{\ell \in \mathbb{Z}} \sum_{Q_\ell^* \in B_\ell^*} \sum_{Q \subset Q_\ell^*} \omega(Q)^{\frac{1}{q}} 2^{k\alpha} \mathbf{D}_k(h)(x_Q) \omega(Q)^{\frac{1}{q'}} 2^{-k\alpha} \mathbf{D}_k^M(g)(x_Q) \right| \\
 &\leq \sum_{\ell \in \mathbb{Z}} \sum_{Q_\ell^* \in B_\ell^*} \left(\sum_{Q \subset Q_\ell^*} \omega(Q) |2^{k\alpha} \mathbf{D}_k(h)(x_Q)|^q \right)^{\frac{1}{q}} \left(\sum_{Q \subset Q_\ell^*} \omega(Q) |2^{-k\alpha} \mathbf{D}_k^M(g)(x_Q)|^{q'} \right)^{\frac{1}{q'}} \\
 &\leq \left\{ \sum_{\ell \in \mathbb{Z}} \sum_{Q_\ell^* \in B_\ell^*} \left(\sum_{Q \subset Q_\ell^*} \omega(Q) |2^{k\alpha} \mathbf{D}_k(h)(x_Q)|^q \right)^{\frac{p}{q}} \left(\sum_{Q \subset Q_\ell^*} \omega(Q) |2^{-k\alpha} \mathbf{D}_k^M(g)(x_Q)|^{q'} \right)^{\frac{p}{q'}} \right\}^{\frac{1}{p}} \\
 &= \left\{ \sum_{\ell \in \mathbb{Z}} \sum_{Q_\ell^* \in B_\ell^*} \left(\sum_{Q \subset Q_\ell^*} \omega(Q) |2^{k\alpha} \mathbf{D}_k(h)(x_Q)|^q \right)^{\frac{p}{q}} \right. \\
 &\quad \left. \times \omega(Q_\ell^*)^{1-\frac{p}{q}} \left(\frac{1}{\omega(Q_\ell^*)^{\frac{q'}{p}-\frac{q'}{q}}} \sum_{Q \subset Q_\ell^*} \omega(Q) |2^{-k\alpha} \mathbf{D}_k^M(g)(x_Q)|^{q'} \right)^{\frac{p}{q'}} \right\}^{\frac{1}{p}} \\
 &\leq \left\{ \sum_{\ell \in \mathbb{Z}} \sum_{Q_\ell^* \in B_\ell^*} \left(\sum_{Q \subset Q_\ell^*} \omega(Q) |2^{k\alpha} \mathbf{D}_k(h)(x_Q)|^q \right)^{\frac{p}{q}} \omega(Q_\ell^*)^{1-\frac{p}{q}} \right\}^{\frac{1}{p}} \|g\|_{CMO_p^{-\alpha, q'}} \\
 &\leq \left\{ \sum_{\ell \in \mathbb{Z}} \left\{ \sum_{Q_\ell^* \in B_\ell^*} \sum_{Q \subset Q_\ell^*} \omega(Q) |2^{k\alpha} \mathbf{D}_k(h)(x_Q)|^q \right\}^{\frac{p}{q}} \left\{ \sum_{Q_\ell^* \in B_\ell^*} \omega(Q_\ell^*) \right\}^{1-\frac{p}{q}} \right\}^{\frac{1}{p}} \|g\|_{CMO_p^{-\alpha, q'}}.
 \end{aligned}$$

To estimate the term $\left\{ \sum_{Q_\ell^* \in B_\ell^*} \omega(Q_\ell^*) \right\}^{1-\frac{p}{q}}$. Set $\tilde{\Omega}_\ell = \left\{ x \in \mathbb{R}^n : \mathbf{M}(\chi_{\Omega_\ell})(x) > \frac{1}{2} \right\}$, where $\mathbf{M}$ is the Hardy–Littlewood maximal function on $\mathbb{R}^n$ with the measure $d\omega$ and $\chi_{\Omega_\ell}(x)$ is the indicate function of Ω_ℓ . It is easily to see that if $Q \in B_\ell$, then $Q \subset \tilde{\Omega}_\ell$. Since all Q_ℓ^* are disjoint, thus

$$\left\{ \sum_{Q_\ell^* \in B_\ell^*} \omega(Q_\ell^*) \right\}^{1-\frac{p}{q}} \leq C \omega(\tilde{\Omega}_\ell)^{1-\frac{p}{q}} \leq C \omega(\Omega_\ell)^{1-\frac{p}{q}},$$

where the first inequality follows from the fact that $\cup_{Q_\ell^* \in B_\ell^*} Q_\ell^* \subset \tilde{\Omega}_\ell$ and $\sum_{Q_\ell^* \in B_\ell^*} \omega(Q_\ell^*) \leq \omega(\tilde{\Omega}_\ell)$, and by the L^2 -boundedness of the Hardy–Littlewood maximal function, the last inequality follows from the estimate $\omega(\tilde{\Omega}_\ell) \leq C \omega(\Omega_\ell)$.

We claim that

$$\sum_{Q_\ell^* \in B_\ell^*} \sum_{Q \subset Q_\ell^*} \omega(Q) \left| 2^{k\alpha} \mathbf{D}_k(h)(x_Q) \right|^q \leq C 2^{q\ell} \omega(\Omega_\ell). \quad (4.2)$$

Under this claim (4.2), we get

$$\begin{aligned} |\langle f, g \rangle| &\leq C \left(\sum_{\ell \in \mathbb{Z}} (2^{q\ell} \omega(\Omega_\ell))^{\frac{p}{q}} \omega(\Omega_\ell)^{1-\frac{p}{q}} \right)^{\frac{1}{p}} \|g\|_{CMO_p^{-a,q'}} \\ &\leq C \left(\sum_{\ell \in \mathbb{Z}} 2^{p\ell} \omega(\Omega_\ell) \right)^{\frac{1}{p}} \|g\|_{CMO_p^{-a,q'}} \\ &\leq C \|\mathbf{S}_q^\alpha(h)\|_p \|g\|_{CMO_p^{-a,q'}} \\ &\leq C \|f\|_{\dot{F}_p^{\alpha,q}} \|g\|_{CMO_p^{-a,q'}}. \end{aligned}$$

It remains to show the claim (4.2). In order to do that, we begin with the following estimate

$$\int_{\tilde{\Omega}_\ell \setminus \Omega_{\ell+1}} \mathbf{S}_q^\alpha(h)(x)^q d\omega(x) \leq C 2^{q\ell} \omega(\tilde{\Omega}_\ell) \leq C 2^{q\ell} \omega(\Omega_\ell).$$

Note that

$$\int_{\tilde{\Omega}_\ell \setminus \Omega_{\ell+1}} \left| \mathbf{S}_q^\alpha(h)(x) \right|^q d\omega(x) \geq \sum_{Q \in B_\ell} \left(2^{k\alpha} \left| \mathbf{D}_k(h)(x_Q) \right| \right)^q \omega\left((\tilde{\Omega}_\ell \setminus \Omega_{\ell+1}) \cap Q \right).$$

Since for each $Q \in B_\ell$ implies $Q \subseteq \tilde{\Omega}_\ell$ and $\Omega_{\ell+1} \subset \Omega_\ell$. Thus

$$\omega\left((\tilde{\Omega}_\ell \setminus \Omega_{\ell+1}) \cap Q \right) = \omega(Q) - \omega(\Omega_{\ell+1} \cap Q) \geq \frac{1}{2} \omega(Q).$$

Therefore,

$$\int_{\tilde{\Omega}_\ell \setminus \Omega_{\ell+1}} \left| \mathbf{S}_q^\alpha(h)(x) \right|^q d\omega(x) \geq C \sum_{Q \in B_\ell} \omega(Q) \left(2^{k\alpha} \left| \mathbf{D}_k(h)(x_Q) \right| \right)^q.$$

This implies the claim (4.2).

The prove of Theorem 2.6(B) is complete. $\square$

Proof of Theorem 2.6(C) Recall Definition 2.4, for $g \in L^2(\mathbb{R}^n, \omega)$, the norm of $g \in \dot{F}_p^{\alpha, \infty}(\mathbb{R}^n, \omega)$ is defined by

$$\|f\|_{\dot{F}_p^{\alpha, \infty}} = \left\| \sup_{k \in \mathbb{Z}, Q \in Q_d^k} 2^{k\alpha} \left| \mathbf{D}_k^M(f)(x_Q) \right| \chi_Q(x) \right\|_{L^p(\mathbb{R}^n, \omega)}.$$

By Hölder's inequality, we have

$$\begin{aligned}
 |\langle f, g \rangle| &= \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) \mathbf{D}_k^M(g)(x_Q) \mathbf{D}_k(h)(x_Q) \right| \\
 &\leq C \int_{\mathbb{R}^n} \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k\alpha} |\mathbf{D}_k(h)(x_Q)| \chi_Q(x) \mathbf{S}_\infty^{-\alpha}(g)(x) d\omega(x) \\
 &\leq C \|\mathbf{S}_1^\alpha(h)\|_p \|\mathbf{S}_\infty^{-\alpha}(g)\|_{p'} \\
 &\leq C \|f\|_{\dot{F}_p^{\alpha,1}} \|g\|_{\dot{F}_{p'}^{-\alpha,\infty}}.
 \end{aligned}$$

To get Theorem 2.6(C), it suffices to show $\|f\|_{\dot{F}_p^{\alpha,1}} \leq C \|f\|_{\dot{F}_p^{\alpha,q}}$.

According to the q -inequality with $q \leq 1$, we find

$$\begin{aligned}
 \|f\|_{\dot{F}_p^{\alpha,1}} &= C \left\{ \int_{\mathbb{R}^n} \left(\sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k\alpha} |\mathbf{D}_k(h)(x_Q)| \chi_Q(x) \right)^p d\omega(x) \right\}^{\frac{1}{p}} \\
 &\leq C \left\{ \int_{\mathbb{R}^n} \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{k\alpha} |\mathbf{D}_k(h)(x_Q)| \right)^q \chi_Q(x) \right|^{\frac{p}{q}} d\omega(x) \right\}^{\frac{1}{p}} \\
 &\leq C \|\mathbf{S}_q^\alpha(h)\|_p \\
 &\leq C \|f\|_{\dot{F}_p^{\alpha,q}}.
 \end{aligned}$$

As a result,

$$|\langle f, g \rangle| \leq C \|f\|_{\dot{F}_p^{\alpha,q}} \|g\|_{\dot{F}_{p'}^{-\alpha,\infty}}.$$

□

Proof of Theorem 2.6(D) Recall Definition 2.5, for $g \in L^2(\mathbb{R}^n, \omega)$, the norm of $g \in \dot{F}_\infty^{\alpha,\infty}(\mathbb{R}^n, \omega)$ is defined by

$$\|g\|_{\dot{F}_\infty^{\alpha,\infty}} = \sup_{k \in \mathbb{Z}, Q \in Q_d^k} 2^{k\alpha} |\mathbf{D}_k^M(g)(x_Q)|.$$

Denote $\beta = -\alpha + N(\frac{1}{p} - 1)$, then we have

$$\begin{aligned}
| \langle f, g \rangle | &= \left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) \mathbf{D}_k^M(g)(x_Q) \mathbf{D}_k(h)(x_Q) \right| \\
&\leq C \int_{\mathbb{R}^n} \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{-k\beta} \left| \mathbf{D}_k(h)(x_Q) \right| \chi_Q(x) d\omega(x) \sup_{k \in \mathbb{Z}, Q \in Q_d^k} 2^{k\beta} \left| \mathbf{D}_k^M(g)(x_Q) \right| \\
&= C \|S_1^{-\beta}(h)\|_1 \|g\|_{\dot{F}_\infty^{\beta, \infty}} \\
&\leq C \|f\|_{\dot{F}_1^{\alpha-N(\frac{1}{p}-1), 1}} \|g\|_{\dot{F}_\infty^{-\alpha+N(\frac{1}{p}-1), \infty}}.
\end{aligned}$$

Recall that $\omega(B(x, r)) \geq Cr^N$ deduced by (1.2). Since $p \leq 1$, and $\ell(Q) = 2^{-k-M}$, where M is a fixed constant, then $2^{-kN(\frac{1}{p}-1)} \leq C\omega(B(x, 2^{-k}))^{\frac{1}{p}-1} \sim C\omega(Q)^{\frac{1}{p}-1}$. Similar to the proof of Theorem 2.6(C), it suffices to show $\|f\|_{\dot{F}_1^{\alpha-N(\frac{1}{p}-1), 1}} \leq C\|f\|_{\dot{F}_p^{\alpha, q}}$. By Minkowski's inequality, and p -inequality with $p \leq 1$, we find

$$\begin{aligned}
\|f\|_{\dot{F}_1^{\alpha-N(\frac{1}{p}-1), 1}} &= \int_{\mathbb{R}^n} \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k(\alpha-N(\frac{1}{p}-1))} \left| \mathbf{D}_k(f)(x_Q) \right| \chi_Q(x) d\omega(x) \\
&\leq C \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k\alpha} \omega(Q)^{\frac{1}{p}-1} \left| \mathbf{D}_k(f)(x_Q) \right| \omega(Q) \\
&= C \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k\alpha} \left| \mathbf{D}_k(f)(x_Q) \right| \omega(Q)^{\frac{1}{p}}.
\end{aligned} \tag{4.3}$$

If $\frac{p}{q} \geq 1$, applying Minkowski's inequality and p -inequality again, the right-hand side of (4.3) is further controlled by

$$\begin{aligned}
&\left(\sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k\alpha p} \left| \mathbf{D}_k(f)(x_Q) \right|^p \omega(Q) \right)^{\frac{1}{p}} \\
&\leq \left\{ \int_{\mathbb{R}^n} \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{k\alpha q} \left| \mathbf{D}_k(f)(x_Q) \right|^q \chi_Q(x) \right)^{\frac{p}{q}} d\omega(x) \right\}^{\frac{1}{p}} \\
&\leq \left\{ \int_{\mathbb{R}^n} \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k\alpha q} \left| \mathbf{D}_k(f)(x_Q) \right|^q \chi_Q(x) \right\}^{\frac{p}{q}} d\omega(x) \right\}^{\frac{1}{p}} \\
&= C \|\mathbf{S}_q^\alpha(f)\|_p \\
&= C \|f\|_{\dot{F}_p^{\alpha, q}}.
\end{aligned}$$

If $\frac{p}{q} < 1$, applying q -inequality and the Minkowski's inequality again, the estimates of right-hand side of (4.3) are following

$$\begin{aligned}
 & \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k\alpha} |\mathbf{D}_k(f)(x_Q)| \omega(Q)^{\frac{1}{p}} \\
 &= \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{k\alpha p} |\mathbf{D}_k(f)(x_Q)|^p \omega(Q) \right)^{\frac{1}{p}} \\
 &\leq \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{k\alpha p} |\mathbf{D}_k(f)(x_Q)|^p \omega(Q) \right)^{\frac{q}{p}} \right\}^{\frac{1}{q}} \\
 &= \left\{ \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(\int_{\mathbb{R}^n} 2^{k\alpha p} |\mathbf{D}_k(f)(x_Q)|^p \chi_Q(x) d\omega(x) \right)^{\frac{q}{p}} \right\}^{\frac{p}{q}} \right\}^{\frac{1}{p}} \\
 &\leq \left\{ \int_{\mathbb{R}^n} \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} 2^{k\alpha q} |\mathbf{D}_k(f)(x_Q)|^q \chi_Q(x) \right\}^{\frac{p}{q}} d\omega(x) \right\}^{\frac{1}{p}} \\
 &= \|\mathbf{S}_q^\alpha(f)\|_p \\
 &= \|f\|_{\dot{F}_p^{\alpha,q}}.
 \end{aligned}$$

The proof of Theorem 2.6(D) is complete. □

The Theorem 2.6(A) indicates that if $\{f_n\}_{n=1}^\infty$ is a sequence in L^2 with $\|\mathbf{S}_q^\alpha(f_n - f_m)\|_p \rightarrow 0$ as $n, m \rightarrow \infty$. Then for each $g \in L^2$ with $\|g\|_{\dot{F}_{p'}^{-\alpha,q'}} < \infty, 1 < p < \infty, 1 < q < \infty$, we have $\lim_{n,m \rightarrow \infty} \langle f_n - f_m, g \rangle = 0$. Therefore, there exists f , as a distribution on $L^2 \cap \dot{F}_{p'}^{-\alpha,q'}$, such that for each $g \in L^2$ with $\|g\|_{\dot{F}_{p'}^{-\alpha,q'}} < \infty$,

$$\langle f, g \rangle = \lim_{n \rightarrow \infty} \langle f_n, g \rangle.$$

Other ranges of p, q stated above have the same results.

5 Dunkl–Triebel–Lizorkin space

In this section, we define the Dunkl–Triebel–Lizorkin spaces, and show the spaces are complete. Before introducing the Dunkl–Triebel–Lizorkin space, we need the following

Lemma 5.1

(A) For $1 < p < \infty, 1 < q < \infty$. Then $D_k(\cdot, y)$ is in $L^2 \cap \dot{F}_{p'}^{-\alpha, q'}$ for any fixed k and $y \in \mathbb{R}^n$. Moreover

$$\|\mathbf{S}_{q'}^{-\alpha}(D_k(\cdot, y))\|_{p'} \leq C,$$

where the constant C only depends on k .

(B) For $\max\left\{\frac{N}{N+1}, \frac{N}{N+\alpha+1}\right\} < p \leq 1, 1 < q < \infty$. Then $D_k(\cdot, y)$ is in $L^2 \cap CMO_p^{-\alpha, q'}$ for any fixed k and $y \in \mathbb{R}^n$. Moreover

$$\sup_P \left(\frac{1}{\omega(P)^{\frac{q'}{p} - \frac{q'}{q}}} \sum_{Q \subset P} \omega(Q) \left(2^{-j\alpha} |\mathbf{D}_j(D_k(\cdot, y))(x_Q)| \right)^{q'} \right)^{\frac{1}{q'}} \leq C$$

where both P and Q are dyadic cubes on $\mathbb{R}^n$ and the constant C which depends on k but is independent of y .

(C) For $1 < p < \infty, \max\left\{\frac{N}{N+1}, \frac{N}{N+\alpha+1}\right\} < q \leq 1$. Then $D_k(\cdot, y)$ is in $L^2 \cap \dot{F}_{p'}^{-\alpha, \infty}$ for any fixed k and $y \in \mathbb{R}^n$. Moreover

$$\|\mathbf{S}_{\infty}^{-\alpha}(D_k(\cdot, y))\|_{p'} = \left\| \sup_{j \in \mathbb{Z}, Q \in \mathcal{Q}_d} 2^{-j\alpha} |\mathbf{D}_j(D_k(\cdot, y))(x_Q)| \chi_Q(x) \right\|_{p'} < C.$$

where the constant C only depends on k .

(D) For $\max\left\{\frac{N}{N+1}, \frac{N}{N+\alpha+1}\right\} < p \leq 1, \max\left\{\frac{N}{N+1}, \frac{N}{N+\alpha+1}\right\} < q \leq 1$. Then $D_k(\cdot, y)$ is in $L^2 \cap \dot{F}_{\infty}^{-\alpha+N(\frac{1}{p}-1), \infty}$ for any fixed k and $y \in \mathbb{R}^n$. Moreover

$$\sup_{j \in \mathbb{Z}, Q \in \mathcal{Q}_d} 2^{(-\alpha+N(\frac{1}{p}-1))j} |\mathbf{D}_j(D_k(\cdot, y))(x_Q)| \leq C.$$

Proof of Lemma 5.1(A) According to Lemma 3.2, we choose $|\alpha| < \varepsilon < 1$ such that

$$|\mathbf{D}_j(D_k(\cdot, y))(x_Q)| \leq C 2^{-|j-k|\varepsilon} \frac{1}{V(x_Q, y, 2^{-jV-k} + d(x_Q, y))} \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x_Q - y\|} \right)^{\varepsilon}.$$

Then,

$$\begin{aligned}
 & \|S_{q'}^{-\alpha}(D_k(\cdot, y))\|_{p'} \\
 &= \left\{ \int_{\mathbb{R}^n} \left\{ \sum_{j \in \mathbb{Z}} \sum_{Q \in Q'_d} \left(2^{-j\alpha} |\mathbf{D}_j(D_k(\cdot, y))(x_Q)| \chi_Q(x) \right)^{q'} \right\}^{\frac{p'}{q'}} d\omega(x) \right\}^{\frac{1}{p'}} \\
 &\leq C \left\{ \int_{\mathbb{R}^n} \left\{ \sum_{j \in \mathbb{Z}} \left(\sum_{Q \in Q'_d} 2^{-j\alpha} 2^{-|j-k|\varepsilon} \right. \right. \right. \\
 &\quad \times \left. \frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\varepsilon \chi_Q(x) \right)^{q'} \left. \right\}^{\frac{p'}{q'}} d\omega(x) \left. \right\}^{\frac{1}{p'}}
 \end{aligned}$$

where the inequality also follows from the fact that $q' > 1$ and $x \in Q$.

If $\frac{p'}{q'} \geq 1$, by Minkowski's inequality, we have

$$\begin{aligned}
 & \|S_{q'}^{-\alpha}(D_k(\cdot, y))\|_{p'} \\
 &\leq C \left\{ \sum_{j \in \mathbb{Z}} \left\{ \int_{\mathbb{R}^n} \left(2^{-j\alpha} 2^{-|j-k|\varepsilon} \frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \right. \right. \right. \\
 &\quad \times \left. \left. \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\varepsilon \right)^{p'} d\omega(x) \right\}^{\frac{q'}{p'}} \left. \right\}^{\frac{1}{q'}} \\
 &\leq C \sum_{j \in \mathbb{Z}} 2^{-j\alpha} 2^{-|j-k|\varepsilon} \left\{ \int_{\mathbb{R}^n} \left(\frac{1}{\omega(x, 2^{-jV-k} + d(x, y))} \right. \right. \\
 &\quad \times \left. \left. \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\varepsilon \right)^{p'} d\omega(x) \right\}^{\frac{1}{p'}}.
 \end{aligned}$$

If $\frac{p'}{q'} < 1$, applying Minkowski's inequality again, and $\frac{1}{p}$ -inequality with $p' > 1$, we have

$$\begin{aligned}
& \|S_{q'}^{-\alpha}(D_k(\cdot, y))\|_{p'} \\
& \leq C \left\{ \int_{\mathbb{R}^n} \sum_{j \in \mathbb{Z}} \left(2^{-j\alpha} 2^{-|j-k|\varepsilon} \frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \right. \right. \\
& \quad \left. \left. \times \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\varepsilon \right)^{p'} d\omega(x) \right\}^{\frac{1}{p'}} \\
& \leq C \left\{ \sum_{j \in \mathbb{Z}} \int_{\mathbb{R}^n} \left(2^{-j\alpha} 2^{-|j-k|\varepsilon} \frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \right. \right. \\
& \quad \left. \left. \times \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\varepsilon \right)^{p'} d\omega(x) \right\}^{\frac{1}{p'}} \\
& \leq C \sum_{j \in \mathbb{Z}} 2^{-j\alpha} 2^{-|j-k|\varepsilon} \left\{ \int_{\mathbb{R}^n} \left(\frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \right. \right. \\
& \quad \left. \left. \times \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\varepsilon \right)^{p'} d\omega(x) \right\}^{\frac{1}{p'}}.
\end{aligned}$$

Thus,

$$\begin{aligned}
& \|S_{q'}^{-\alpha}(D_k(\cdot, y))\|_{p'} \\
& \leq C \sum_{j \in \mathbb{Z}} 2^{-j\alpha} 2^{-|j-k|\varepsilon} \left\{ \int_{\mathbb{R}^n} \left(\frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \right. \right. \\
& \quad \left. \left. \times \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\varepsilon \right)^{p'} d\omega(x) \right\}^{\frac{1}{p'}}. \tag{5.1}
\end{aligned}$$

For $j \geq k$,

$$\begin{aligned}
& \int_{\mathbb{R}^n} \left(\frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\varepsilon \right)^{p'} d\omega(x) \\
& \leq C \int_{d(x, y) < 2^{-k}} \left(\frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\varepsilon \right)^{p'} d\omega(x) \\
& \quad + \sum_{m=1}^{\infty} \int_{2^{-k+(m-1)} \leq d(x, y) < 2^{-k+m}} \left(\frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \right. \\
& \quad \left. \times \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\varepsilon \right)^{p'} d\omega(x).
\end{aligned}$$

Note that

$$\begin{aligned}
 & \int_{d(x,y) < 2^{-k}} \left(\frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\epsilon \right)^{p'} d\omega(x) \\
 & \leq C \sum_{\sigma \in G} \int_{\|\sigma(y) - x\| < 2^{-k}} \frac{1}{\omega(B(x, 2^{-k}))^{p'}} d\omega(x) \\
 & \leq C \sum_{\sigma \in G} \int_{\|\sigma(y) - x\| < 2^{-k}} \frac{1}{\omega(B(\sigma(y), 2^{-k}))^{p'}} d\omega(x) \\
 & \leq C \omega(B(\sigma(y), 2^{-k}))^{1-p'} \\
 & \leq C 2^{-kN(1-p')}.
 \end{aligned}$$

And for $m \geq 1$,

$$\begin{aligned}
 & \int_{2^{-k+(m-1)} \leq d(x,y) < 2^{-k+m}} \left(\frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \right. \\
 & \quad \times \left. \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\epsilon \right)^{p'} d\omega(x) \\
 & \leq C 2^{-m\epsilon} \sum_{\sigma \in G} \int_{\|\sigma(y) - x\| < 2^{-k+m}} \frac{1}{\omega(B(\sigma(y), 2^{-k+m}))^{p'}} d\omega(x) \\
 & \leq C 2^{-m(\epsilon+(p'-1)N)} 2^{-kN(1-p')}.
 \end{aligned}$$

Thus, for $j \geq k$,

$$\int_{\mathbb{R}^n} \left(\frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\epsilon \right)^{p'} d\omega(x) \leq C 2^{-kN(1-p')} \leq C_k,$$

where C_k is a constant depending on k .

Similarly, for $j < k$,

$$\int_{\mathbb{R}^n} \left(\frac{1}{\omega(B(x, 2^{-jV-k} + d(x, y)))} \left(\frac{2^{-jV-k}}{2^{-jV-k} + \|x - y\|} \right)^\epsilon \right)^{p'} d\omega(x) \leq C 2^{-jN(1-p')}.$$

Together with (5.1) and $|\alpha| < \epsilon$, we have

$$\|S_{q'}^{-\alpha}(D_k(\cdot, y))\|_{p'} \leq C_k \sum_{j: j \geq k} 2^{-j\alpha} 2^{-(j-k)\epsilon} + \sum_{j: j < k} 2^{-j\alpha} 2^{(j-k)\epsilon} 2^{-jN(\frac{1}{p'}-1)} \leq C_k.$$

□

Proof of Lemma 5.1(B) Fix a dyadic cube P with the side length 2^{-M-j_0} and the center x_P . For $Q \in \mathcal{Q}_d^j$, applying Lemma 3.2, we have

$$\begin{aligned}
& \left| \mathbf{D}_j(D_k(\cdot, y))(x_Q) \right| \\
& \leq C 2^{-|j-k|\varepsilon} \frac{1}{V(x_Q, y, 2^{-j\nu-k} + d(x_Q, y))} \left(\frac{2^{-j\nu-k}}{2^{-j\nu-k} + \|x_Q - y\|} \right)^\varepsilon \\
& \leq C 2^{-|j-k|\varepsilon} \frac{1}{\omega(B(x_Q, 2^{-j\nu-k}))} \\
& \leq C 2^{-|j-k|\varepsilon} \frac{1}{\omega(B(x_Q, 2^{-k}))}.
\end{aligned} \tag{5.2}$$

We have the estimate

$$\begin{aligned}
& \frac{1}{\omega(P)^{\frac{q'}{p} - \frac{q'}{q}}} \sum_{Q \subset P} \omega(Q) \left(2^{-j\alpha} \left| \mathbf{D}_j(D_k(\cdot, y))(x_Q) \right| \right)^{q'} \\
& = \frac{1}{\omega(P)^{\frac{q'}{p} - \frac{q'}{q}}} \sum_{j=j_0}^{\infty} \sum_{\{Q \in Q_d^j : Q \subset P\}} \omega(Q) \left(2^{-j\alpha} \left| \mathbf{D}_j(D_k(\cdot, y))(x_Q) \right| \right)^{q'} \\
& \leq C \frac{1}{\omega(P)^{\frac{q'}{p} - \frac{q'}{q}}} \sum_{j=j_0}^{\infty} \sum_{\{Q \in Q_d^j : Q \subset P\}} \omega(Q) 2^{-j\alpha q'} 2^{-|j-k|\varepsilon q'} \frac{1}{\omega(B(x_Q, 2^{-k}))^{q'}} \\
& \leq C \sup_{x \in P} \frac{1}{\omega(B(x, 2^{-k}))^{q'}} \frac{1}{\omega(P)^{\frac{q'}{p} - \frac{q'}{q}}} \sum_{j=j_0}^{\infty} \sum_{\{Q \in Q_d^j : Q \subset P\}} \omega(Q) 2^{-(j\alpha + |j-k|\varepsilon)q'} \\
& \leq C \sup_{x \in P} \frac{1}{\omega(B(x, 2^{-k}))^{q'}} \frac{1}{\omega(P)^{\frac{q'}{p} - \frac{q'}{q}}} \sum_{j=j_0}^{\infty} \omega(P) 2^{-(j\alpha + |j-k|\varepsilon)q'} \\
& \leq C \sup_{x \in P} \frac{1}{\omega(B(x, 2^{-k}))^{q'}} \frac{1}{\omega(P)^{\frac{q'}{p} - \frac{q'}{q} - 1}} \sum_{j=j_0}^{\infty} 2^{-(j\alpha + |j-k|\varepsilon)q'}.
\end{aligned}$$

(i) For $j_0 \geq k$, the doubling property of the measure ω implies

$$\omega(B(x_P, 2^{-k})) \leq C 2^{(-k+j_0)N} \omega(B(x_P, 2^{-j_0})) \sim C 2^{(-k+j_0)N} \omega(P).$$

Thus, for $|\alpha| < \varepsilon$,

$$\begin{aligned}
& \frac{1}{\omega(P)^{\frac{q'}{p} - \frac{q'}{q}}} \sum_{Q \subset P} \omega(Q) \left(2^{-j\alpha} \left| \mathbf{D}_j(D_k(\cdot, y))(x_Q) \right| \right)^{q'} \\
& \leq C \sup_{x \in P} \frac{1}{\omega(B(x, 2^{-k}))^{q'}} \frac{1}{\omega(B(x_P, 2^{-k}))^{\frac{q'}{p} - \frac{q'}{q} - 1}} 2^{(-k+j_0)N(\frac{q'}{p} - \frac{q'}{q} - 1)} \sum_{j=j_0}^{\infty} 2^{-(j-k)\varepsilon + j\alpha q'} \\
& \leq C_k \sup_{x \in P} \frac{1}{\omega(B(x, 2^{-k}))^{\frac{q'}{p}}}.
\end{aligned}$$

(ii) For $j_0 < k$, since

$$\omega(B(x_Q, 2^{-k})) \leq \omega(B(x_Q, 2^{-j_0})) \sim \omega(P),$$

we have

$$\begin{aligned} & \frac{1}{\omega(P)^{\frac{q'_p - q'_q}{p}}} \sum_{Q \subset P} \omega(Q) \left(2^{-j\alpha} \left| \mathbf{D}_j(D_k(\cdot, y))(x_Q) \right| \right)^{q'} \\ & \leq C \sup_{x \in P} \frac{1}{\omega(B(x, 2^{-k}))^{q'}} \frac{1}{\omega(P)^{\frac{q'_p - q'_q}{p} - 1}} \sum_{j=j_0}^{\infty} 2^{-(j\alpha + |j-k|\varepsilon)q'} \\ & \leq C_k \sup_{x \in P} \frac{1}{\omega(B(x, 2^{-k}))^{\frac{q'_p}{p}}}. \end{aligned}$$

Taking the supremum over all dyadic cubes P and using $\inf_{x \in \mathbb{R}^n} \omega(B(x, 1)) \geq C$, we obtain

$$\sup_P \left(\frac{1}{\omega(P)^{\frac{q'_p - q'_q}{p}}} \sum_{Q \subset P} \omega(Q) \left(2^{-j\alpha} \left| \mathbf{D}_j(D_k(\cdot, x_Q))(y) \right| \right)^{q'} \right)^{\frac{1}{q'}} \leq C_k$$

and the proof of Lemma 5.1(B) is complete. $\square$

Proof of Lemma 5.1(C)

$$\begin{aligned} & \|S_{\infty}^{-\alpha}(D_k(\cdot, y))\|_{p'} \\ & = \left\{ \int_{\mathbb{R}^n} \left| \sup_{j \in \mathbb{Z}, Q \in Q_d^j} 2^{-j\alpha} \left| \mathbf{D}_j(D_k(\cdot, y))(x_Q) \right| \chi_Q(x) \right|^{p'} d\omega(x) \right\}^{\frac{1}{p'}} \\ & \leq C \left\{ \int_{\mathbb{R}^n} \left| \sup_{j \in \mathbb{Z}, Q \in Q_d^j} 2^{-j\alpha} 2^{-|j-k|\varepsilon} \right. \right. \\ & \quad \times \left. \frac{1}{V(x_Q, y, 2^{-j\nu-k} + d(x_Q, y))} \left(\frac{2^{-j\nu-k}}{2^{-j\nu-k} + \|x_Q - y\|} \right)^{\varepsilon} \chi_Q(x) \right|^{p'} d\omega(x) \left. \right\}^{\frac{1}{p'}} \\ & \leq C \left\{ \sup_{j \in \mathbb{Z}, Q \in Q_d^j} 2^{-j\alpha p'} 2^{-|j-k|\varepsilon p'} \int_Q \left(\frac{1}{\omega(B(x, 2^{-j\nu-k} + d(x, y)))} \right. \right. \\ & \quad \times \left. \left. \left(\frac{2^{-j\nu-k}}{2^{-j\nu-k} + \|x - y\|} \right)^{\varepsilon} \right)^{p'} d\omega(x) \right\}^{\frac{1}{p'}} \\ & \leq C \sup_{j \in \mathbb{Z}} 2^{-j\alpha} 2^{-|j-k|\varepsilon} \left\{ \int_{\mathbb{R}^n} \left(\frac{1}{\omega(B(x, 2^{-j\nu-k} + d(x, y)))} \right. \right. \\ & \quad \times \left. \left. \left(\frac{2^{-j\nu-k}}{2^{-j\nu-k} + \|x - y\|} \right)^{\varepsilon} \right)^{p'} d\omega(x) \right\}^{\frac{1}{p'}}. \end{aligned}$$

Similar to the proof of Lemma 5.1(A), we have

$$\|S_{\infty}^{-\alpha}(D_k(\cdot, y))\|_{p'} \leq C_k \sup_{j: j \geq k} 2^{-j\alpha} 2^{-(j-k)\varepsilon} + \sup_{j: j < k} 2^{-j\alpha} 2^{(j-k)\varepsilon} 2^{-jN(\frac{1}{p}-1)} \leq C_k.$$

□

Proof of Lemma 5.1(D) Recall (5.2), we have

$$|\mathbf{D}_j(D_k(\cdot, y))(x_Q)| \leq C 2^{-|j-k|\varepsilon} \frac{1}{\omega(B(x_Q, 2^{-k}))}.$$

(i) For $j \geq k$, since $p > \frac{N}{N+\alpha+\varepsilon}$, then

$$\begin{aligned} & \sup_{j: j \geq k; Q \in Q_d^j} 2^{(-\alpha+N(\frac{1}{p}-1))j} |\mathbf{D}_j(D_k(\cdot, y))(x_Q)| \\ & \leq C \sup_{j: j \geq k; Q \in Q_d^j} 2^{(-\alpha+N(\frac{1}{p}-1))j} 2^{-(j-k)\varepsilon} \frac{1}{\omega(B(x_Q, 2^{-k}))} \\ & \leq C_k, \end{aligned}$$

(ii) For $j < k$, since $p \leq 1$ and $|\alpha| < \varepsilon$, then

$$\begin{aligned} & \sup_{j: j < k; Q \in Q_d^j} 2^{(-\alpha+N(\frac{1}{p}-1))j} |\mathbf{D}_j(D_k(\cdot, y))(x_Q)| \\ & \leq C \sup_{j: j < k; Q \in Q_d^j} 2^{(-\alpha+N(\frac{1}{p}-1))j} 2^{(j-k)\varepsilon} \frac{1}{\omega(B(x_Q, 2^{-k}))} \\ & \leq C_k. \end{aligned}$$

□

The proof of Lemma 5.1 is complete.

For $1 < p < \infty, 1 < q < \infty$, we denote $L^2 \cap \dot{F}_{p'}^{-\alpha, q'}$, as the subspace of $f \in L^2$, with the norm $\|f\|_{\dot{F}_{p'}^{-\alpha, q'}} < \infty$. Based on the above Lemma 5.1, if $f \in \left(L^2 \cap \dot{F}_{p'}^{-\alpha, q'}\right)'$, then $\mathbf{D}_k(f)(x)$ is well defined since for each fixed x , $D_k(x, y) \in L^2 \cap \dot{F}_{p'}^{-\alpha, q'}$. Other ranges of p, q have similar results.

Theorem 2.7 describes an important property for each distribution f . More precisely, it establishes the weak-type discrete Calderón reproducing formula in the distribution sense. Now we prove Theorem 2.7.

Proof of Theorem 2.7(A) By the Theorem 2.6, there exists $f \in \left(L^2 \cap \dot{F}_{p'}^{-\alpha, q'}\right)'$ such that for each $g \in L^2 \cap \dot{F}_{p'}^{-\alpha, q'}$,

$$\langle f, g \rangle = \lim_{n \rightarrow \infty} \langle f_n, g \rangle.$$

Observing that

$$\|\mathbf{S}_q^\alpha(f - f_n)\|_p = \|\mathbf{S}_q^\alpha(\lim_{m \rightarrow \infty} (f_m - f_n))\|_p \leq \liminf_{m \rightarrow \infty} \|\mathbf{S}_q^\alpha(f_m - f_n)\|_p,$$

hence $\|\mathbf{S}_q^\alpha(f - f_n)\|_p \rightarrow 0$ as $n \rightarrow \infty$. This implies that

$$\|f\|_{\dot{F}_p^{\alpha,q}} = \|\mathbf{S}_q^\alpha(f)\|_p = \lim_{n \rightarrow \infty} \|\mathbf{S}_q^\alpha(f_n)\|_p = \lim_{n \rightarrow \infty} \|f_n\|_{\dot{F}_p^{\alpha,q}} < \infty.$$

Applying Theorem 2.2, for each f_n there exists an h_n such that $\|f_n\|_2 \sim \|h_n\|_2$ and $\|f_n\|_{\dot{F}_p^{\alpha,q}} \sim \|h_n\|_{\dot{F}_p^{\alpha,q}}$. Thus by Theorem 2.6, there exists $h \in \left(L^2 \cap \dot{F}_{p'}^{-\alpha,q'}\right)'$ such that for each $g \in L^2 \cap \dot{F}_{p'}^{-\alpha,q'}$, we have

$$\langle h, g \rangle = \lim_{n \rightarrow \infty} \langle h_n, g \rangle.$$

Therefore,

$$\|\mathbf{S}_q^\alpha(h_n - h_m)\|_p \rightarrow 0,$$

and

$$\|h\|_{\dot{F}_p^{\alpha,q}} = \|\mathbf{S}_q^\alpha(h)\|_p = \lim_{n \rightarrow \infty} \|\mathbf{S}_q^\alpha(h_n)\|_p \sim \lim_{n \rightarrow \infty} \|\mathbf{S}_q^\alpha(f_n)\|_p = \|\mathbf{S}_q^\alpha(f)\|_p = \|f\|_{\dot{F}_p^{\alpha,q}}.$$

For each $g \in L^2 \cap \dot{F}_{p'}^{-\alpha,q'}$, we know that

$$\left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) \mathbf{D}_k^M(g)(x_Q) \mathbf{D}_k(h)(x_Q) \right| \leq C \|f\|_{\dot{F}_p^{\alpha,q}} \|g\|_{\dot{F}_{p'}^{-\alpha,q'}},$$

which implies that the series $\sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(x, x_Q) \mathbf{D}_k(h)(x_Q)$ is a distribution in $\left(L^2 \cap \dot{F}_{p'}^{-\alpha,q'}\right)'$. Moreover, by the reproducing formula of f_n in Theorem 2.2, for each $g \in L^2 \cap \dot{F}_{p'}^{-\alpha,q'}$,

$$\langle f, g \rangle = \lim_{n \rightarrow \infty} \langle f_n, g \rangle = \lim_{n \rightarrow \infty} \left\langle \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(\cdot, x_Q) \mathbf{D}_k(h_n)(x_Q), g(\cdot) \right\rangle,$$

where $\|f_n\|_2 \sim \|h_n\|_2$ and $\|f_n\|_{\dot{F}_p^{\alpha,q}} \sim \|h_n\|_{\dot{F}_p^{\alpha,q}}$.

By the same proof of Theorem 2.6, we have

$$\left| \left\langle \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(\cdot, x_Q) \mathbf{D}_k(h - h_n)(x_Q), g(\cdot) \right\rangle \right| \leq C \|h_n - h\|_{\dot{F}_p^{\alpha,q}} \|g\|_{\dot{F}_{p'}^{-\alpha,q'}}.$$

Since $\|h_n - h\|_{\dot{F}_p^{\alpha,q}} \rightarrow 0$, as $n \rightarrow \infty$, we have

$$\langle f, g \rangle = \lim_{n \rightarrow \infty} \langle f_n, g \rangle = \left\langle \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(\cdot, x_Q) \mathbf{D}_k(h)(x_Q), g(\cdot) \right\rangle.$$

□

The proof of Theorem 2.7(B), (C), (D) is similar, so we omit the details.

The Theorem 2.7(A) indicates that one can consider $L^2 \cap \dot{F}_{p'}^{-\alpha,q'}$, $|\alpha| < 1$, $1 < p < \infty$, $1 < q < \infty$, the subspace of $f \in L^2$ with the norm $\|f\|_{\dot{F}_{p'}^{-\alpha,q'}} < \infty$, as the test function space and $(L^2 \cap \dot{F}_{p'}^{-\alpha,q'})'$ as the distribution space. The Dunkl–Triebel–Lizorkin space is defined by Definition 2.8. We remark that in the Definition 2.8, the series

$$\sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) \lambda_Q D_k^M(x, x_Q) \text{ with } \left\| \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} (2^{k\alpha} |\lambda_Q|)^q \chi_Q \right\} \right\|_p^{\frac{1}{q}} < \infty$$

defines a distribution in $(L^2 \cap \dot{F}_{p'}^{-\alpha,q'})'$. Indeed, applying the proof of Theorem 2.6 for each $g \in L^2 \cap \dot{F}_{p'}^{-\alpha,q'}$,

$$\left| \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) \lambda_Q \mathbf{D}_k^M(g)(x_Q) \right| \leq C \left\| \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} (2^{k\alpha} |\lambda_Q|)^q \chi_Q \right\} \right\|_p^{\frac{1}{q}} \|g\|_{\dot{F}_{p'}^{-\alpha,q'}}.$$

Other ranges of p , q has the same results.

Proof of Theorem 2.9(A) Suppose $f \in \dot{F}_p^{\alpha,q}(\mathbb{R}^n, \omega)$. Then $f \in (L^2 \cap \dot{F}_{p'}^{-\alpha,q'})'$ and f has a wavelet-type decomposition $f(x) = \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) \lambda_Q D_k^M(x, x_Q)$ in $(L^2 \cap \dot{F}_{p'}^{-\alpha,q'})'$ with its norm $\left\| \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} (2^{k\alpha} |\lambda_Q|)^q \chi_Q(x) \right\} \right\|_p^{\frac{1}{q}} < \infty$. Set

$$f_n(x) = \sum_{|k| \leq n} \sum_{\substack{Q \in Q_d^k \\ Q \subseteq B(0, n)}} \omega(Q) \lambda_Q D_k^M(x, x_Q).$$

Then $f_n \in L^2 \cap \dot{F}_p^{\alpha,q}$ and f_n converges to f in $(L^2 \cap \dot{F}_{p'}^{-\alpha,q'})'$ as $n \rightarrow \infty$.

To see that $f \in \overline{L^2 \cap \dot{F}_p^{\alpha,q}}$, by Theorem 2.7, it suffices to show that $\|f_n - f_m\|_{\dot{F}_p^{\alpha,q}} \rightarrow 0$ as $n, m \rightarrow \infty$. Indeed, if let $E_n = \{(k, Q) : |k| \leq n, Q \in Q_d^k \subseteq B(0, n)\}$ and $E_{n,m}^c = E_n \setminus E_m$ with $n \geq m$,

$$\begin{aligned} & \|f_n - f_m\|_{\dot{F}_p^{\alpha,q}} \\ &= \left\| \left(\sum_{k' \in \mathbb{Z}} \sum_{Q' \in Q_d^k} \left(2^{k'\alpha} \left| \mathbf{D}_{k'}^M(f_n - f_m)(x_{Q'}) \right| \right)^q \chi_{Q'}(x) \right)^{\frac{1}{q}} \right\|_p \\ &\leq \left\| \left(\sum_{k' \in \mathbb{Z}} \sum_{Q' \in Q_d^k} \left(2^{k'\alpha} \left| \mathbf{D}_{k'}^M \left(\sum_{E_{n,m}^c} \omega(Q) \lambda_Q D_k^M(\cdot, x_Q) \right) (x_{Q'}) \right| \right)^q \chi_{Q'}(x) \right)^{\frac{1}{q}} \right\|_p \\ &\leq C \left\| \left\{ \sum_{E_{n,m}^c} \left(2^{k\alpha} \left| \lambda_Q \right| \right)^q \chi_Q(x) \right\}^{\frac{1}{q}} \right\|_p \rightarrow 0, \end{aligned}$$

as $n, m \rightarrow \infty$, where the last inequality follows from the same proof of **Step 1** in the Theorem 2.2 and hence, $f \in \overline{L^2 \cap \dot{F}_p^{\alpha,q}}$.

Conversely, if $f \in \overline{L^2 \cap \dot{F}_p^{\alpha,q}}$ by Theorem 2.6, then there exists $h \in \left(L^2 \cap \dot{F}_{p'}^{-\alpha,q'} \right)'$ with $\|S_q^\alpha(h)\|_p \sim \|S_q^\alpha(f)\|_p$ such that for each $g \in L^2 \cap \dot{F}_{p'}^{-\alpha,q'}$,

$$\langle f, g \rangle = \left\langle \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) D_k^M(\cdot, x_Q) \mathbf{D}_k(h)(x_Q), g(\cdot) \right\rangle.$$

Set $\lambda_Q = \mathbf{D}_k(h)(x_Q)$ with $Q \in Q_d^k$. We obtain a wavelet-type decomposition of f in $\left(L^2 \cap \dot{F}_{p'}^{-\alpha,q'} \right)'$ in the distribution sense:

$$f(x) = \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \omega(Q) \lambda_Q D_k^M(x, x_Q).$$

Hence, $f \in \dot{F}_p^{\alpha,q}(\mathbb{R}^n, \omega)$. Moreover

$$\|f\|_{\dot{F}_p^{\alpha,q}} = \inf \left\{ \left\| \left\{ \sum_{k \in \mathbb{Z}} \sum_{Q \in Q_d^k} \left(2^{k\alpha} \left| \lambda_Q \right| \right)^q \chi_Q(x) \right\}^{\frac{1}{q}} \right\|_p \right\} \leq C \|S_q^\alpha(h)\|_p \leq C \|S_q^\alpha(f)\|_p.$$

□

The proof of Theorem 2.9(B), (C), (D) are similar, so we omit the details.

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